Guidelines

Early Detection of Prostate Cancer: AUA/SUO Guideline (2026)

PUBLISHED 2023; AMENDED 2026

Unabridged version of this Guideline [pdf]

Algorithm associated with this Guideline [pdf]

Amendment summary for this Guideline [pdf]

To cite this guideline:

Lin DW, Carlsson S, Filson CP, et al. Updates to Early Detection of Prostate Cancer: AUA/SUO Guideline (2026).J Urol. 0(0).10.1097/JU.0000000000004995. https://www.auajournals.org/doi/10.1097/JU.0000000000004995

Panel Members

John T. Wei, MD, MS; Daniel Barocas, MD; Sigrid Carlsson, MD, PhD, MPH; Fergus Coakley, MD; Scott Eggener, MD; Ruth Etzioni, PhD; Samson W. Fine, MD; Misop Han, MD; Badrinath R. Konety, MD; Martin Miner, MD; Kelvin Moses, MD, PhD; Merel G. Nissenberg, JD; Peter A. Pinto, MD; Simpa S. Salami, MD, MPH; Lesley Souter, PhD; Ian M. Thompson, MD; Daniel W. Lin, MD

Amendment Panel

Daniel W. Lin, MD; Sigrid Carlsson, MD, PhD, MPH; Christopher P. Filson, MD, MS; Badrinath R. Konety, MD; Andrei S. Purysko, MD

Staff and Consultants

Sennett K. Kim; Erin Kirkby, MS; Lesley H. Souter, PhD

SUMMARY

Purpose

The recommendations discussed on the early detection of prostate cancer provide a framework to facilitate clinical decision-making in the implementation of prostate cancer screening and follow-up.

Methodology

The systematic review of this Guideline was based on searches in Ovid MEDLINE and Embase and Cochrane Database of Systematic Reviews (January 1, 2000 – November 21, 2022). Searches were supplemented by reviewing reference lists of relevant articles. Criteria for inclusion and exclusion of studies were based on the Key Questions (KQs) and the populations, interventions, comparators, outcomes, timing, types of studies and settings (PICOTS) of interest. The target population was persons without a diagnosis of prostate cancer undergoing prostate-specific antigen (PSA) screening, or patients without prostate cancer who have a suspicious finding indicating possible clinically significant prostate cancer who are undergoing or considering an initial or repeat biopsy. In 2025, the Early Detection of Prostate Cancer Guideline was updated through the American Urological Association (AUA) amendment process. This process involved reviewing and integrating newly published literature into previously established Guidelines. The methodologist updated the original Guideline search strategy to systematically search Ovid MEDLINE and Embase for new evidence published between November 2022 and December 2024.

GUIDELINE STATEMENTS

PSA Screening

1. Clinicians should engage in shared decision-making (SDM) with people for whom prostate cancer screening would be appropriate and proceed based on a person’s values and preferences. (*Clinical Principle*)
2. When screening for prostate cancer, clinicians should use PSA as the first screening test. (*Strong Recommendation; Evidence Level: Grade A*)
3. For people with a newly elevated PSA, clinicians should repeat the PSA prior to a secondary biomarker, imaging, or biopsy. (*Expert Opinion*)
4. Clinicians may begin prostate cancer screening and offer a baseline PSA test to people between ages 45 to 50 years. (*Conditional Recommendation; Evidence Level: Grade B*)
5. Clinicians should offer prostate cancer screening beginning at age 40 to 45 years for people at increased risk of developing prostate cancer based on the following: Black race, germline mutations, strong family history of prostate cancer. (*Strong Recommendation; Evidence Level: Grade B*)
6. Clinicians should offer regular prostate cancer screening every 2 to 4 years to people aged 50 to 69 years. (*Strong Recommendation; Evidence Level: Grade A*)
7. Clinicians may personalize the re-screening interval, or decide to discontinue screening, based on patient preference, age, PSA, prostate cancer risk, life expectancy, and general health following SDM. (*Conditional Recommendation; Evidence Level: Grade B*)
8. Clinicians may use digital rectal examination (DRE) alongside PSA to establish risk of clinically significant prostate cancer. (*Conditional Recommendation; Evidence Level: Grade C*)
9. For people undergoing prostate cancer screening, clinicians should not use PSA velocity as the sole indication for secondary biomarker, imaging, or biopsy. (*Strong Recommendation; Evidence Level: Grade B*)

10. Clinicians and patients may use validated risk calculators to inform the SDM process regarding prostate biopsy. (*Conditional Recommendation; Evidence Level: Grade B*)

11. When the risk of clinically significant prostate cancer is sufficiently low based on available clinical, laboratory, and imaging data, clinicians and patients may forgo near-term prostate biopsy. (*Clinical Principle*)

Initial Biopsy

12. Clinicians should inform patients undergoing a prostate biopsy that there is a risk of identifying a cancer, with a sufficiently low risk of mortality, that could safely be monitored with active surveillance (AS) rather than treated. (*Clinical Principle*)

13. Clinicians may use magnetic resonance imaging (MRI) prior to initial biopsy to increase the detection of Grade Group (GG)2+ prostate cancer. (*Conditional Recommendation*; *Evidence Level: Grade A*)

14. Radiologists should utilize Prostate Imaging Reporting and Data System (PI-RADS) in the reporting of multi-parametric magnetic resonance imaging (mpMRI). (*Moderate Recommendation; Evidence Level: Grade C*)

15. For biopsy-naïve patients who have a suspicious lesion on MRI, clinicians should perform targeted biopsies of the suspicious lesion and may also perform a systematic template biopsy. (*Moderate Recommendation [targeted biopsies]/Conditional Recommendation [systematic template biopsy]; Evidence Level: Grade C)*

16. For patients with both absence of suspicious findings on MRI and elevated risk for GG2+ prostate cancer, clinicians should proceed with a systematic biopsy. (*Moderate Recommendation; Evidence Level: Grade C*)

17. Clinicians may use adjunctive urine or serum markers when further risk stratification would influence the decision regarding whether to proceed with biopsy. (*Conditional Recommendation*; *Evidence Level: Grade C*)

18. For patients with a PSA > 50 ng/mL and no clinical concerns for infection or other cause for increased PSA (e.g., recent prostate instrumentation), clinicians may omit a prostate biopsy in cases where biopsy poses significant risk or where the need for prostate cancer treatment is urgent (e.g., impending spinal cord compression). (*Expert Opinion*)

Repeat Biopsy

19. Clinicians should communicate with patients following biopsy to review biopsy results, reassess risk of undetected or future development of GG2+ disease, and mutually decide whether to discontinue screening, continue screening, or perform adjunctive testing for early reassessment of risk. (*Clinical Principle*)

20. Clinicians should not discontinue prostate cancer screening based solely on a negative prostate biopsy. (*Strong Recommendation; Evidence Level: Grade C*)

21. After a negative biopsy, clinicians should not solely use a PSA threshold to decide whether to repeat the biopsy. (*Strong Recommendation; Evidence Level: Grade B*)

22. If the clinician and patient decide to continue screening after a negative biopsy, clinicians should re-evaluate the patient within the normal screening interval (two to four years) or sooner, depending on risk of clinically significant prostate cancer and life expectancy. (*Clinical Principle*)

23. At the time of re-evaluation after negative biopsy, clinicians should use a risk assessment tool that incorporates the protective effect of prior negative biopsy. (*Strong Recommendation; Evidence Level: Grade B*)

24. After a negative initial biopsy in patients with low probability for harboring GG2+ prostate cancer, clinicians should not reflexively perform biomarker testing. (*Clinical Principle*)

25. After a negative biopsy, clinicians may use blood-, urine-, or tissue-based biomarkers selectively for further risk stratification if results are likely to influence the decision regarding repeat biopsy or otherwise substantively change the patient's management. (*Conditional Recommendation; Evidence Level: Grade C*)

26. In patients with focal (one core) high-grade prostatic intraepithelial neoplasia (HGPIN) on biopsy, clinicians should not perform immediate repeat biopsy. (*Moderate Recommendation; Evidence Level: Grade C*)

27. In patients with multifocal HGPIN, clinicians may proceed with additional risk evaluation, guided by PSA/DRE anc mpMRI findings. (*Expert Opinion*)

28. In patients with atypical small acinar proliferation (ASAP), clinicians should perform additional testing, which may include repeat biopsy. (*Moderate Recommendation; Evidence Level: Grade C*)

29. In patients with atypical intraductal proliferation (AIP), clinicians should perform additional testing. (*Expert Opinion*)

30. In patients undergoing repeat biopsy with no prior prostate MRI, clinicians should obtain a prostate MRI prior to biopsy. (*Strong Recommendation; Evidence Level: Grade C*)

31. In patients with indications for a repeat biopsy who do not have a suspicious lesion on MRI, clinicians may proceed with a systematic biopsy. (*Conditional Recommendation; Evidence Level: Grade B*)

32. In patients undergoing repeat biopsy and who have a suspicious lesion on MRI, clinicians should perform targeted biopsies of the suspicious lesion and may also perform a systematic template biopsy. (*Moderate Recommendation [targeted biopsies]/Conditional Recommendation [systematic template biopsy]; Evidence Level: Grade C)*

Biopsy Technique

33. Clinicians may use software registration of MRI and ultrasound images during fusion biopsy, when available. (*Expert Opinion*)

34. Clinicians should obtain at least two needle biopsy cores per target in patients with suspicious prostate lesion(s) on MRI. (*Moderate Recommendation; Evidence Level: Grade C*)

35. Clinicians may use either a transrectal or transperineal biopsy route when performing a biopsy. (*Conditional Recommendation; Evidence Level: Grade B*)

INTRODUCTION

Background

Prostate cancer is the most commonly diagnosed noncutaneous malignancy in American men. It is estimated that 333,830 patients will be diagnosed with prostate cancer, and 36,320 deaths from prostate cancer will occur in the United States (U.S.) in 2026.[1] There was an estimated 1,414,259 new cases of prostate cancer and 375,304 deaths worldwide in 2020.[2] Significant advances have been made in early detection, especially with the increasing availability and usage of biomarkers and mpMRI. This Guideline is based on a systematic review of the recently published literature and addresses early detection with an emphasis on PSA-based screening, considerations for initial and repeat biopsy, and biopsy technique, with the goal of identifying clinically significant prostate cancer.

Terminology and Definitions

This Guideline provides recommendations for prostate cancer screening in different groups based on their age range and risk criteria, with an emphasis on SDM. SDM is particularly necessary as there is no universally accepted standard definition of low versus elevated risk for prostate cancer detection. In practice, clinicians often resort to an elevated PSA level based on laboratory, prostate size, or age-based "norms" as a surrogate for an elevated prostate cancer risk, but such definitions, while easy to apply, do not suffice for all people and circumstances. Thus, clinicians may tailor the definitions of elevated risk and elevated PSA to the clinical situation at hand. Some examples that may elevate risk of clinically significant prostate cancer are Black race, germline mutations, strong family history of prostate cancer, polygenic risk scores (numerical estimate of an individual's genetic predisposition for developing a specific condition [e.g., prostate cancer]),[3, 4] and other factors that may be indicated by risk calculators (e.g., total PSA, PSA density, percent free PSA, age). More importantly, this Guideline emphasizes potential benefit in using validated risk calculators and provides recommendations for the timing and methodology for screening.

This Guideline underscores the goal of detecting "clinically significant" cancer for initial and repeat biopsy. The risk of mortality in patients with GG1 prostate cancer is extremely low.[5, 6] Thus, this Guideline defines clinically significant prostate cancer as GG2 or higher (GG2+) prostate cancer and will use "clinically significant prostate cancer" and "GG2+" interchangeably throughout. However, the Panel acknowledges there are various definitions of "clinically significant" as not all "clinically significant" cancers are destined to impact quality or quantity-of-life, and it is patient-specific. The Guideline recommends utilizing validated risk calculators, particularly calculators that incorporate previous negative biopsy and mpMRI use in the repeat biopsy setting. It also addresses the significance of non-cancerous, yet potentially significant, pathologic findings identified from the biopsy. With the emergence of mpMRI and novel biomarkers, the Panel evalua
the current evidence to develop recommendations on how best to incorporate these into clinical practice. In certain cl

scenarios, additional data are needed to make definitive recommendations for the optimal biopsy approach. An abnormal MRI, for the purpose of this Guideline, is defined as PI-RADS 3 to 5 as supported by much of the literature. However, given the local variation and expertise in reading MRIs, some clinicians may opt to limit an abnormal MRI to PI-RADS 4 to 5.

This Guideline is intended for all patient populations with a prostate gland. For consistency purposes, this Guideline refers to these individuals as "people" or "patients" throughout this document.

METHODOLOGY

The systematic review utilized to inform this Guideline was conducted by an independent methodological consultant. Determination of the Guideline scope and review of the final systematic review to inform Guideline statements was conducted in conjunction with the Early Detection of Prostate Cancer Panel. In 2025, an independent methodologist conducted a search for studies published between November 2022 and December 2024 to capture new evidence published since the last update review. A second search that incorporated keywords for newly developed biomarkers was conducted in Ovid MEDLINE and Embase and identified studies published from January 2000 through December 2024 to align with the search inception date used in the original Guideline.

Panel Formation

The Panel was created in 2021 by the American Urological Association Education and Research, Inc. (AUAER). The Practice Guidelines Committee (PGC) of the AUA selected the Panel Chairs who in turn appointed the additional panel members with specific expertise in this area. The multidisciplinary panel includes representation from urology/urologic oncology, epidemiology, biostatistics, primary care, pathology, and radiology. The Panel additionally included patient representation. Funding of the Panel was provided by the AUA; panel members received no remuneration for their work.

The Early Detection of Prostate Cancer Amendment Panel was established in 2025 by the AUA to review new literature and provide updates herein.

Searches and Article Selection

A search was conducted for existing systematic reviews on October 11, 2021 and updated on November 21, 2022. Systematic reviews published as a component of practice Guidelines were also considered eligible for inclusion. An electronic search employing Ovid was used to systematically search the MEDLINE and Embase databases, as well as the Cochrane Library, for systematic reviews evaluating detection of prostate cancer.

When systematic reviews were not identified, or when identified reviews were incomplete, Ovid was used to systematically search MEDLINE and Embase databases for articles evaluating detection of prostate cancer utilizing the PICOTS elements. During PICOTS development, panel members submitted landmark studies addressing the KQs to the methodologist. These studies were defined as control articles and were compared with the literature search strategy output; the strategy was subsequently updated as necessary to capture all control articles. Databases were originally searched for studies published from January 1, 2000 through October 11, 2021 and subsequently updated to November 21, 2022. In addition to the MEDLINE and Embase database searches, reference lists of included systematic reviews and primary literature were scanned for potentially useful studies.

All hits from the Ovid literature search were input into a reference management software (EndNote X7), where duplicate citations were removed. Abstracts were reviewed by the methodologist to determine if each study addressed the KQs and met study design inclusion criteria. For all research questions, randomized controlled trials (RCTs), observational studies, modelling studies with theoretical cohorts, and case-control studies were considered for inclusion in the evidence base. For all KQs, studies had to enroll at least 30 patients per study arm. Case series, letters, editorials, *in vitro* studies, studies conducted in animal models, and studies not published in English were excluded from the evidence base *a priori*.

Full-text review was conducted on studies that passed the abstract screening phase. Studies were compared to the PICOTS criteria. Ten panel members were paired with the methodologist and completed duplicate full-text study selection of 10% of studies undergoing full-text review. The dual-review trained the methodologist, who then completed full-time review of the remaining studies.

In 2024, the methodologist conducted a literature search using Ovid MEDLINE and Embase for new evidence published between November 2022 and December 2024. A second search that incorporated keywords for newly developed markers was conducted in the same databases and identified studies published from January 2000 through December 2024 to align with the search inception date used in the original Guideline. The literature search returned 15,193 citations after deduplication. Following the study selection process outlined for the original Guideline and using the original PICOTS criteria, 40 new studies were added to the evidence base.

Titles and abstracts of studies identified by the search were reviewed in a two-stage process in Rayyan.[7] During the first stage, studies were reviewed to determine if they assessed early-detection of prostate cancer, and if they met the study selection criteria of prespecified study type. Studies not published in English were excluded from the evidence base. Allowable study types included systematic reviews, RCTs, observational studies, modeling studies with theoretical cohorts, and case-control studies. All other study types were excluded.

During the second stage of title and abstract review, abstracts were compared to the PICOTS criteria. Additionally, during the second stage of review, studies were assessed to determine if they could reaffirm or refute the original Guideline statements. In alignment with the Guideline, for all KQs, studies had to enroll at least 30 patients per study arm. Additionally, for studies reporting on detection rates of prostate cancer, at least 50 prostate cancers had to be detected for inclusion in the evidence base. This size limitation was waived for one of the KQs due to a paucity of studies.

Full-text review was conducted on studies that passed the title and abstract screen and flagged by the Panel to potentially change Guideline statements or inform new Guideline statements. Complete studies were compared to the PICOTS criteria. Those that met the PICOTS criteria and informed the proposed Guideline statement changes were included in the evidence base. Studies that did not inform the statements were defined as those with reported outcomes that did not affirm nor refute the proposed changes.

Data Abstraction

Data were extracted from all studies that passed full-text review by the methodologist.

Risk of Bias Assessment

Quality assessment for all retained studies was conducted. Using this method, studies deemed to be of low quality would not be excluded from the systematic review, but would be retained, and their methodological strengths and weaknesses were discussed where relevant. To evaluate the risk of bias within the identified studies, the Assessment of Multiple Systematic Reviews (AMSTAR),[8] tool was used for systematic reviews, the Cochrane Risk of Bias Tool[9] was used for randomized studies, a Risk of Bias in Non-Randomized Studies of Intervention (ROBINS-I)[10] was used for observational studies and modeling studies with theoretical cohorts, and Quality Assessment of Diagnostic Accuracy Studies-2 (QUADAS-2)[11] was used for diagnostic accuracy studies. Additional important quality features, such as comparison type, power of statistical analysis, and sources of funding were extracted for each study.

Data Synthesis

Meta-analysis was appropriate for studies informing four KQs and six outcomes using RevMan.[12] For all meta-analyses there was substantial heterogeneity in both the patient populations and the methodologies employed within the studies, making random-effects methods the most appropriate. Odds ratios for detection of clinically significant prostate cancer using MRI-targeted biopsy alone and fusion biopsy plus systematic biopsy were calculated based on raw data reported in studies and pooled using an inverse-variance method. For calculation of the number of avoided biopsies and missed clinically significant prostate cancer using various biomarkers in both biopsy naïve and repeat biopsy populations, prevalence and standard errors were extracted or calculated from reported raw data in studies and pooled using an inverse variance method. Finally, prevalence and standard errors for clinically significant prostate cancer detection using a PI-RADS score of 1 to 2, 3, 4, and 5 were calculated from raw data reported in studies and pooled using an inverse-variance method.[12] Due to the paucity of data using only PI-RADS version 2.1, pooled studies used version 1.0 through version 2.1.

Determination of Evidence Strength

The Grading of Recommendations Assessment, Development, and Evaluation (GRADE)[13] system was used to determine the aggregate evidence quality for each outcome, or group of related outcomes, informing KQs. GRADE defines a body of evidence in relation to how confident Guideline developers can be that the estimate of effects as reported by that body of evidence, is correct. Evidence is categorized as high, moderate, low, and very low, and assessment is based on the aggregate risk of bias for the evidence base, plus limitations introduced as a consequence of inconsistency, indirectness, imprecision, and publication bias across the studies.[14] Additionally, certainty of evidence can be downgraded if confounding across the studies has resulted in the potential for the evidence base to overestimate the effect. Upgrading of evidence is possible if the body of evidence indicates a large effect or if confounding would suggest either spurious effects or would reduce the demonstrated effect.

The AUA employs a 3-tiered strength of evidence system to underpin evidence-based Guideline statements. **Table 1** summarizes the GRADE categories, definitions, and how these categories translate to the AUA strength of evidence categories. In short, high certainty by GRADE translates to AUA A-category strength of evidence, moderate to B, and both low and very low to C.

The AUA categorizes body of evidence strength as Grade A (e.g., well-conducted and highly-generalizable RCTs or exceptionally strong observational studies with consistent findings), Grade B (e.g., RCTs with some weaknesses of procedure or generalizability or moderately strong observational studies with consistent findings), or Grade C (e.g., RCTs with serious deficiencies of procedure or generalizability or extremely small sample sizes or observational studies that are inconsistent, have small sample sizes, or have other problems that potentially confound interpretation of data). By definition, Grade A evidence is evidence about which the Panel has a high level of certainty, Grade B evidence is evidence about which the Panel has a moderate level of certainty, and Grade C evidence is evidence about which the Panel has a low level of certainty.[15]

Table 1: Strength of Evidence Definitions

AUA Strength of Evidence Category	GRADE Certainty Rating	Definition
A	High	• Very confident that the true effect lies close to that of the estimate of the effect
B	Moderate	• Moderately confident in the effect estimate • The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different
C	Low	• Confidence in the effect estimate is limited • The true effect may be substantially different from the estimate of the effect
	Very Low	• Very little confidence in the effect estimate • The true effect is likely to be substantially different from the estimate of effect

AUA Nomenclature: Linking Statement Type to Evidence Strength

The AUA nomenclature system explicitly links statement type to body of evidence strength, level of certainty, magnitude of benefit or risk/burdens, and the Panel's judgment regarding the balance between benefits and risks/burdens (**Table 2**). Strong Recommendations are directive statements that an action should (benefits outweigh risks/burdens) or should not (risks/burdens outweigh benefits) be undertaken because net benefit or net harm is substantial. Moderate Recommendations are directive statements that an action should (benefits outweigh risks/burdens) or should not (risks/burdens outweigh benefits) be undertaken because net benefit or net harm is moderate. Conditional Recommendations are non-directive statements used when the evidence indicates that there is no apparent net benefit or harm or when the balance between benefits and risks/burden is unclear. All three statement types may be supported by any body of evidence strength grade. Body of evidence strength Grade A in support of a Strong or Moderate Recommendation indicates that the statement can be applied to most patients in most circumstances and future research is unlikely to change confidence. Body of evidence strength Grade B in support of a Strong or Moderate Recommendation indicates that the statement can be applied to most patients in most circumstances, but better evidence could change confidence. Body of evidence strength Grade C in support of a Strong or Moderate Recommendation indicates that the statement can be applied to most patients in most circumstances, but better evidence is likely to change confidence. Conditional Recommendations also can be supported by any evidence strength. When body of evidence strength is Grade A, the statement indicates that benefits and risks/burdens appear balanced, the best action depends on patient circumstances, and future research is unlikely to change confidence. When body of evidence strength Grade B is used, benefits and risks/burdens appear balanced, the best action also depends on individual patient circumstances and better evidence could change confidence. When body of evidence strength Grade C is used, there is uncertainty regarding the balance between benefits and risks/burdens, alternative strategies may be equally reasonable, and better evidence is likely to change confidence.

Where gaps in the evidence existed, the Panel provides guidance in the form of Clinical Principles or Expert Opinions with consensus achieved using a modified Delphi technique if differences in opinion emerged.[16] A Clinical Principle is a statement about a component of clinical care that is widely agreed upon by urologists or other clinicians for which there may or may not be evidence in the medical literature. Expert Opinion refers to a statement, achieved by consensus of the Panel, that is based on members' clinical training, experience, knowledge, and judgment.

Table 2: AUA Nomenclature Linking Statement Type to Level of Certainty, Magnitude of Benefit or Risk/Burden, and Body of Evidence Strength

Evidence Grade	Evidence Strength A (High Certainty)	Evidence Strength B (Moderate Certainty)	Evidence Strength C (Low Certainty)
Strong Recommendation (Net benefit or harm substantial)	-Benefits > Risks/Burdens (or vice versa) -Net benefit (or net harm) is substantial -Applies to most patients in most circumstances and future research is unlikely to change confidence	-Benefits > Risks/Burdens (or vice versa) -Net benefit (or net harm) is substantial -Applies to most patients in most circumstances but better evidence could change confidence	-Benefits > Risks/Burdens (or vice versa) -Net benefit (or net harm) appears substantial -Applies to most patients in most circumstances but better evidence is likely to change confidence (rarely used to support a Strong Recommendation)
Moderate Recommendation (Net benefit or harm moderate)	-Benefits > Risks/Burdens (or vice versa) -Net benefit (or net harm) is moderate -Applies to most patients in most circumstances and future research is unlikely to change confidence	-Benefits > Risks/Burdens (or vice versa) -Net benefit (or net harm) is moderate -Applies to most patients in most circumstances but better evidence could change confidence	-Benefits > Risks/Burdens (or vice versa) -Net benefit (or net harm) appears moderate -Applies to most patients in most circumstances but better evidence is likely to change confidence
Conditional Recommendation (Net benefit or harm comparable to other options)	-Benefits = Risks/Burdens -Best action depends on individual patient circumstances -Future Research is unlikely to change confidence	-Benefits = Risks/Burdens -Best action appears to depend on individual patient circumstances -Better evidence could change confidence	-Balance between Benefits & Risks/Burdens unclear -Net benefit (or net harm) comparable to other options -Alternative strategies may be equally reasonable -Better evidence likely to change confidence
Clinical Principle	a statement about a component of clinical care that is widely agreed upon by urologists or other clinicians for which there may or may not be evidence in the medical literature		
Expert Opinion	a statement, achieved by consensus of the Panel, that is based on members' clinical training, experience, knowledge, and judgment for which there may or may not be evidence in the medical literature		

Peer Review and Document Approval

An integral part of the Guideline development process at the AUA is external peer review. The AUA conducted a comprehensive peer review process to ensure that the document was reviewed by experts who were knowledgeable in the area of early detection of prostate cancer. In addition to reviewers from the AUA PGC, Science and Quality Council (SQC), and Board of Directors (BOD), the document was reviewed by external content experts. Additionally, a call for reviewers was placed on the AUA website from October 10 to 24 of 2022 to allow any additional interested parties to request a copy of the document for review. The Guideline was also sent to the Urology Care Foundation and members of the AUA Patient Advocacy network to open the document further to the patient perspective. The draft Guideline document was distributed to 174 peer reviewers. All peer review comments were blinded and sent to the Panel for review. In total, 84 reviewers provided comments, including 69 external reviewers. At the end of the peer review process, a total of 770 comments were received. Following comment discussion, the Panel revised the draft as needed. Once finalized, the Guideline was submitted to the AUA PGC, SQC, and BOD for final approval as well as the Society of Urologic Oncology (SUO).

In 2025, as part of the amendment process, the AUA conducted a comprehensive peer review. A call for reviewers was issued from August 12 to August 29, 2025. On October 2, 2025 the draft Guideline document was distributed to a total of 88 peer reviewers, with 84 submitting comments. The Amendment Panel carefully reviewed and discussed all submitted comments, making revisions to the draft as necessary. Once finalized, the Guideline was submitted for approval to the PGC, SQC, and representatives from SUO. It was subsequently presented to the AUA BOD for final approval.

GUIDELINE STATEMENTS

PSA Screening

GUIDELINE STATEMENT 1

Clinicians should engage in SDM with people for whom prostate cancer screening would be appropriate and proceed based on a person's values and preferences. (*Clinical Principle*)

Discussion

GUIDELINE STATEMENT 2

When screening for prostate cancer, clinicians should use PSA as the first screening test. (***Strong Recommendation; Evidence Level: Grade A***)

Discussion

GUIDELINE STATEMENT 3

For people with a newly elevated PSA, clinicians should repeat the PSA prior to a secondary biomarker, imaging, or biopsy. (***Expert Opinion***)

Discussion

GUIDELINE STATEMENT 4

Clinicians may begin prostate cancer screening and offer a baseline PSA test to people between ages 45 to 50 years. (***Conditional Recommendation; Evidence Level: Grade B***)

Discussion

GUIDELINE STATEMENT 5

Clinicians should offer prostate cancer screening beginning at age 40 to 45 years for people at increased risk of developing prostate cancer based on the following: Black race, germline mutations, strong family history of prostate cancer. (***Strong Recommendation; Evidence Level: Grade B***)

Discussion

GUIDELINE STATEMENT 6

Clinicians should offer regular prostate cancer screening every 2 to 4 years to people aged 50 to 69 years. (***Strong Recommendation; Evidence Level: Grade A***)

Discussion

GUIDELINE STATEMENT 7

Clinicians may personalize the re-screening interval, or decide to discontinue screening, based on patient preference, age, PSA, prostate cancer risk, life expectancy, and general health following SDM. (***Conditional Recommendation; Evidence Level: Grade B***)

Discussion

GUIDELINE STATEMENT 8

Clinicians may use DRE alongside PSA to establish risk of clinically significant prostate cancer. (***Conditional Recommendation; Evidence Level: Grade C***)

Discussion

GUIDELINE STATEMENT 9

For people undergoing prostate cancer screening, clinicians should not use PSA velocity as the sole indication for a secondary biomarker, imaging, or biopsy. (***Strong Recommendation; Evidence Level: Grade B***)

Discussion

GUIDELINE STATEMENT 10

Clinicians and patients may use validated risk calculators to inform the SDM process regarding prostate biopsy. (***Conditional Recommendation; Evidence Level: Grade B***)

Discussion

GUIDELINE STATEMENT 11

When the risk of clinically significant prostate cancer is sufficiently low based on available clinical, laboratory, and imaging data, clinicians and patients may forgo near-term prostate biopsy. (***Clinical Principle***)

Discussion

Initial Biopsy

GUIDELINE STATEMENT 12

Clinicians should inform patients undergoing a prostate biopsy that there is a risk of identifying a cancer, with a sufficiently low risk of mortality, that could safely be monitored with AS rather than treated. (***Clinical Principle***)

Discussion

GUIDELINE STATEMENT 13

Clinicians may use MRI prior to initial biopsy to increase the detection of GG2+ prostate cancer. (***Conditional Recommendation***; ***Evidence Level: Grade A***)

Discussion

GUIDELINE STATEMENT 14

Radiologists should utilize PI-RADS in the reporting of mpMRI. (***Moderate Recommendation; Evidence Level: Grade C***)

Discussion

GUIDELINE STATEMENT 15

For biopsy-naïve patients who have a suspicious lesion on MRI, clinicians should perform targeted biopsies of the suspicious lesion and may also perform a systematic template biopsy. (***Moderate Recommendation [targeted biopsies]***/***Conditional Recommendation [systematic template biopsy]; Evidence Level: Grade C)***

Discussion

GUIDELINE STATEMENT 16

For patients with both absence of suspicious findings on MRI and elevated risk for GG2+ prostate cancer, clinicians should proceed with a systematic biopsy. (***Moderate Recommendation; Evidence Level: Grade C***)

Discussion

GUIDELINE STATEMENT 17

Clinicians may use adjunctive urine or serum markers when further risk stratification would influence the decision regarding whether to proceed with biopsy. (***Conditional Recommendation***; ***Evidence Level: Grade C***)

Discussion

GUIDELINE STATEMENT 18

For patients with a PSA > 50 ng/mL and no clinical concerns for infection or other cause for increased PSA (e.g., recent prostate instrumentation), clinicians may omit a prostate biopsy in cases where biopsy poses significant risk or where the need for prostate cancer treatment is urgent (e.g., impending spinal cord compression). (***Expert Opinion***)

Discussion

Repeat Biopsy

GUIDELINE STATEMENT 19

Clinicians should communicate with patients following biopsy to review biopsy results, reassess risk of undetected or future development of GG2+ disease, and mutually decide whether to discontinue screening, continue screening, or perform adjunctive testing for early reassessment of risk. (***Clinical Principle***)

GUIDELINE STATEMENT 20

Clinicians should not discontinue prostate cancer screening based solely on a negative prostate biopsy. (***Strong Recommendation; Evidence Level: Grade C***)

GUIDELINE STATEMENT 21

After a negative biopsy, clinicians should not solely use a PSA threshold to decide whether to repeat the biopsy. (***Strong Recommendation; Evidence Level: Grade B***)

GUIDELINE STATEMENT 22

If the clinician and patient decide to continue screening after a negative biopsy, clinicians should re-evaluate the patient within the normal screening interval (two to four years) or sooner, depending on risk of clinically significant prostate cancer and life expectancy. (***Clinical Principle***)

GUIDELINE STATEMENT 23

At the time of re-evaluation after negative biopsy, clinicians should use a risk assessment tool that incorporates the protective effect of prior negative biopsy. (***Strong Recommendation; Evidence Level: Grade B***)

Discussion

GUIDELINE STATEMENT 24

After a negative initial biopsy in patients with low probability for harboring GG2+ prostate cancer, clinicians should not reflexively perform biomarker testing. (***Clinical Principle***)

Discussion

GUIDELINE STATEMENT 25

After a negative biopsy, clinicians may use blood-, urine-, or tissue-based biomarkers selectively for further risk stratific
if results are likely to influence the decision regarding repeat biopsy or otherwise substantively change the patient's management. (***Conditional Recommendation; Evidence Level: Grade C***)

Discussion

GUIDELINE STATEMENT 26

In patients with focal (one core) HGPIN on biopsy, clinicians should not perform immediate repeat biopsy. (***Moderate Recommendation; Evidence Level: Grade C***)

Discussion

GUIDELINE STATEMENT 27

In patients with multifocal HGPIN, clinicians may proceed with additional risk evaluation, guided by PSA/DRE and mpMRI findings. (***Expert Opinion***)

Discussion

GUIDELINE STATEMENT 28

In patients with ASAP, clinicians should perform additional testing, which may include repeat biopsy. (***Moderate Recommendation; Evidence Level: Grade C***)

Discussion

GUIDELINE STATEMENT 29

In patients with AIP, clinicians should perform additional testing. (***Expert Opinion***)

Discussion

GUIDELINE STATEMENT 30

In patients undergoing repeat biopsy with no prior prostate MRI, clinicians should obtain a prostate MRI prior to biopsy. (***Strong Recommendation; Evidence Level: Grade C***)

Discussion

GUIDELINE STATEMENT 31

In patients with indications for a repeat biopsy who do not have a suspicious lesion on MRI, clinicians may proceed with a systematic biopsy. (***Conditional Recommendation; Evidence Level: Grade B***)

Discussion

GUIDELINE STATEMENT 32

In patients undergoing repeat biopsy and who have a suspicious lesion on MRI, clinicians should perform targeted biopsies of the suspicious lesion and may also perform a systematic template biopsy. (***Moderate Recommendation [targeted biopsies]/Conditional Recommendation [systematic template biopsy]; Evidence Level: Grade C)***

Discussion

Biopsy Technique

GUIDELINE STATEMENT 33

Clinicians may use software registration of MRI and ultrasound images during fusion biopsy, when available. (*Expert Opinion*)

Discussion

GUIDELINE STATEMENT 34

Clinicians should obtain at least two needle biopsy cores per target in patients with suspicious prostate lesion(s) on MRI. (*Moderate Recommendation; Evidence Level: Grade C*)

Discussion

GUIDELINE STATEMENT 35

Clinicians may use either a transrectal or transperineal biopsy route when performing a biopsy. (*Conditional Recommendation; Evidence Level: Grade B*)

Discussion

FUTURE DIRECTIONS

Screening and diagnosis of prostate cancer remain intensely debated topics with major implications for individual and population health. There continue to be many unanswered questions that can prompt future research, preferably in the form of clinical trials and modeling studies to enhance and optimize patient care. Future trials will hopefully prioritize inclusion of historically underrepresented populations.

SDM regarding whether to screen, how frequently, and when to proceed to secondary testing (e.g., imaging or biomarkers) or biopsy is critically important. However, clinicians tend to discuss potential benefits of screening far more frequently than potential harms.[327] There is an unmet need for decision aids in multiple languages for persons at various levels of health literacy which clearly and comprehensively inform the patient of potential benefits and harms.

For populations at higher risk of being diagnosed with prostate cancer, such as those with a concerning family history of prostate cancer, Black race, genetic risk, or elevated baseline PSA, a targeted and perhaps more intensive screening warrants further investigation and is the focus of ongoing clinical trials (e.g., NCT04472338, NCT05129605). Additionally, investigation of novel approaches is strongly encouraged which may have operating characteristics which outperform currently available tools. Conversely, to minimize overdetection rates, people with a very low likelihood of clinically significant prostate cancer may benefit from less intensive or discontinuation of screening.

Although emerging data exist, a far more comprehensive understanding is required of the impact of race and ethnicity on the operating characteristics of PSA, secondary biomarkers, and prostate imaging. It is also essential to recognize many people undergoing screening are of mixed (or unknown) race and ethnicity. Since dramatic disparities exist regarding access and affordability of certain diagnostic or imaging modalities, efforts should be made by clinicians, payors, and health care systems to bridge this gap.

For non-binary patients or transgender women there is a lack of data on prostate cancer screening preferences, if and when to initiate, the accuracy of biomarkers (e.g., PSA, secondary biomarkers, MRI), potential psychological consequences, impact of gender-affirming hormonal therapy, and priorities regarding management options.[328] Considerably more effort and research are required.

While there are a plethora of serum, urine, tissue, and imaging biomarkers to assess the likelihood of high-grade prostate cancer, there is little knowledge on comparative effectiveness, how they may complement or supplement each other, and how various stepwise algorithms perform. Considerable research is required to achieve the goal of a highly effective, practical, scalable, and widely available approach.

While this update extensively reviewed emerging data surrounding MRI imaging of the prostate, other imaging technologies, such as micro-ultrasound, have shown similar promise in the detection of clinically significant prostate cancer in patients with elevated PSA.[329] Furthermore, no imaging technique has been shown to impact meaningful long-term outcomes such as cancer-specific mortality. Even with growing clinical experience with mpMRI and fusion biopsies, there remain some cases concerning GG2+ cancer where the targeted biopsy either did not detect cancer or only detected GG1 disease. While this may be due to false positive mpMRI reading, it is also possible that the lesion was under-sampled (e.g., small target in a difficult to access location), and the use of perilesional + lesion-only biopsies is being investigated in retrospective data. On a practical level, the impact of interobserver variability and reliance on high-quality imaging and expert interpretation have been highlighted by recent studies. As such, the future directions in the imaging domain will focus on evolving MRI protocols, such as biparametric MRI, and the use of computer-aided and artificial intelligence-enhanced interpretation of MRI-acquired prostate anatomic and radiomic imaging.[121, 330]

TOOLS AND RESOURCES

Prostate Cancer
Filtered Resources

ABBREVIATIONS

95%CI	95% confidence interval
5-ARI	5-alpha reductase inhibitor
ACR	American College of Radiology
AHRQ	Agency of Healthcare Research and Quality
AIP	Atypical intraductal proliferation
AMSTAR	Assessment of Multiple Systematic Reviews
aOR	adjusted odds ratio
AS	Active surveillance
ASAP	Atypical small acinar proliferation
AUA	American Urological Association
AUAER	American Urological Association Education and Research, Inc.
AUC	Area under the curve
BOD	Board of Directors
CDR	Cancer detection rate
DRE	Digital rectal exam
EMR	Electronic medical records
ERSPC	European Randomized Study of Screening for Prostate Cancer
GG	Grade Group
GRADE	Grading of Recommendations Assessment, Development, and Evaluation
HGPIN	High-grade prostatic intraepithelial neoplasia
hK2	Human kallikrein 2
IDC-P	Intraductal carcinoma of prostate
LATP	Local anesthetic transperineal prostate biopsy
MIC-1	Macrophage inhibitory cytokine-1
mpMRI	multi-parametric magnetic resonance imaging
MPS	MyProstateScore
MRI	Magnetic resonance imaging
MSMB	Microseminoprotein-beta
MUSIC	Michigan Urological Surgery Improvement Collaborative
NND	Number needed to diagnose

NNS	Number needed to screen
NPV	Negative predictive value
PBCG	Prostate biopsy collaborative group
PCA3	Prostate Cancer Antigen 3
PCPT	Prostate cancer prevention trial
PGC	Practice Guidelines Committee
PHI	Prostate health index
PICOTS	Populations, interventions, comparators, outcomes, timing, and settings
PI-RADS	Prostate Imaging Reporting & Data System
PLCO	The Prostate, Lung, Colorectal and Ovarian
PPV	Positive predictive value
PRS	Polygenic risk score
PSA	Prostate-specific antigen
QUADAS- 2	Quality Assessment of Diagnostic Accuracy Studies-2
RCT	Randomized controlled trial
ROBINS-I	Risk of Bias in Non-Randomized Studies of Intervention
ROC	Receiver operating characteristic curve
RR	Relative risk
SDM	Shared decision-making
SNP	Single nucleotide polymorphism
SQC	Science and Quality Council
SSA	Social security administration
STHLM-3	Stockholm-3
SUO	Society of Urologic Oncology
TRUS	Transrectal ultrasound
U.S.	United States

REFERENCES

1. Siegel RL, Kratzer TB, Wagle NS et al: Cancer statistics, 2026. CA: A Cancer Journal for Clinicians 2026; **76:** e70043
2. Wang L, Lu B, He M et al: Prostate cancer incidence and mortality: Global status and temporal trends in 89 countries from 2000 to 2019. Front Public Health 2022; **10:** 811044
3. McHugh JK, Bancroft EK, Saunders E et al: Assessment of a polygenic risk score in screening for prostate cancer. N Engl J Med 2025; **392:** 1406
4. Conti DV, Darst BF, Moss LC et al: Trans-ancestry genome-wide association meta-analysis of prostate cancer identifies new susceptibility loci and informs genetic risk prediction. Nat Genet 2021; **53:** 65
5. Eggener SE, Scardino PT, Walsh PC et al: Predicting 15-year prostate cancer specific mortality after radical prostatectomy. J Urol 2011; **185:** 869
6. Mahal BA, Berman RA, Taplin ME and Huang FW: Prostate cancer-specific mortality across gleason scores in black vs nonblack men. Jama 2018; **320:** 2479
7. Ouzzani M, Hammady H, Fedorowicz Z and Elmagarmid A: Rayyan-a web and mobile app for systematic reviews. Syst Rev 2016; **5:** 210
8. Shea BJ, Grimshaw JM, Wells GA et al: Development of amstar: A measurement tool to assess the methodological quality of systematic reviews. BMC Med Res Methodol 2007; **7:** 10
9. Higgins JP, Altman DG, Gotzsche PC et al: The cochrane collaboration's tool for assessing risk of bias in randomised trials. Bmj 2011; **343:** d5928
10. Sterne JA, Hernan MA, Reeves BC et al: Robins-i: A tool for assessing risk of bias in non-randomised studies of interventions. Bmj 2016; **355:** i4919
11. Whiting PF, Rutjes AW, Westwood ME et al: Quadas-2: A revised tool for the quality assessment of diagnostic accuracy studies. Ann Intern Med 2011; **155:** 529

12. The Nordic Cochrane Centre: Review manager (revman), Version 5.3 ed. Copenhagen: The Chochrane Collaboration 2014

13. Guyatt G, Oxman AD, Akl EA et al: Grade guidelines: 1. Introduction-grade evidence profiles and summary of findings tables. J Clin Epidemiol 2011; **64:** 383

14. Balshem H, Helfand M, Schünemann HJ et al: Grade guidelines: 3. Rating the quality of evidence. J Clin Epidemiol 2011; **64:** 401

15. Faraday M, Hubbard H, Kosiak B and Dmochowski R: Staying at the cutting edge: A review and analysis of evidence reporting and grading; the recommendations of the american urological association. BJU Int 2009; **104:** 294

16. Hsu C-C and Sandford B: The delphi technique: Making sense of consensus. Practical Assessment, Research and Evaluation 2007; **12**

17. Makarov DV, Chrouser K, Gore JL et al: Aua white paper on implementation of shared decision making into urological practice. Urology Practice 2016; **3:** 355

18. Stacey D, Legare F, Lewis K et al: Decision aids for people facing health treatment or screening decisions. Cochrane Database Syst Rev 2017; **4:** CD001431

19. Riikonen JM, Guyatt GH, Kilpelainen TP et al: Decision aids for prostate cancer screening choice: A systematic review and meta-analysis. JAMA Intern Med 2019; **179:** 1072

20. Eastham JA, Boorjian SA and Kirkby E: Clinically localized prostate cancer: Aua/astro guideline. J Urol 2022; **208:** 505

21. Hugosson J, Roobol MJ, Mansson M et al: A 16-yr follow-up of the european randomized study of screening for prostate cancer. Eur Urol 2019; **76:** 43

22. Hugosson J, Godtman RA, Carlsson SV et al: Eighteen-year follow-up of the goteborg randomized population-based prostate cancer screening trial: Effect of sociodemographic variables on participation, prostate cancer incidence and mortality. Scand J Urol 2018; **52:** 27

23. Gronberg H, Adolfsson J, Aly M et al: Prostate cancer screening in men aged 50-69 years (sthlm3): A prospective population-based diagnostic study. Lancet Oncol 2015; **16:** 1667

24. Nordstrom T, Discacciati A, Bergman M et al: Prostate cancer screening using a combination of risk-prediction, mri, and targeted prostate biopsies (sthlm3-mri): A prospective, population-based, randomised, open-label, non-inferiority trial. Lancet Oncol 2021; **22:** 1240

25. Eklund M, Jaderling F, Discacciati A et al: Mri-targeted or standard biopsy in prostate cancer screening. New England Journal of Medicine 2021; **385:** 908

26. Eastham JA, Riedel E, Scardino PT et al: Variation of serum prostate-specific antigen levels: An evaluation of year-to-year fluctuations. JAMA 2003; **289:** 2695

27. Nordström T, Adolfsson J, Grönberg H and Eklund M: Repeat prostate-specific antigen tests before prostate biopsy decisions. J Natl Cancer Inst 2016; **108:** djw165

28. Eggener SE, Large MC, Gerber GS et al: Empiric antibiotics for an elevated prostate-specific antigen (psa) level: A randomised, prospective, controlled multi-institutional trial. BJU Int 2013; **112:** 925

29. Greiman A, Shah J, Bhavsar R et al: Six weeks of fluoroquinolone antibiotic therapy for patients with elevated serum prostate-specific antigen is not clinically beneficial: A randomized controlled clinical trial. Urology 2016; **90:** 32

30. Crawford ED, Schutz MJ, Clejan S et al: The effect of digital rectal examination on prostate-specific antigen levels. JAMA 1992; **267:** 2227

31. Jiandani D, Randhawa A, Brown RE et al: The effect of bicycling on psa levels: A systematic review and meta-analysis. Prostate Cancer Prostatic Dis 2015; **18:** 208

32. Tarhan F, Demir K, Orcun A and Madenci OC: Effect of ejaculation on serum prostate-specific antigen concentration. Int Braz J Urol 2016; **42:** 472

33. Gulati R, Gore JL and Etzioni R: Comparative effectiveness of alternative prostate-specific antigen--based prostate cancer screening strategies: Model estimates of potential benefits and harms. Annals of Internal Medicine 2013; **158:**

145

34. Oesterling JE, Jacobsen SJ, Chute CG et al: Serum prostate-specific antigen in a community-based population of healthy men. Establishment of age-specific reference ranges. Jama 1993; **270:** 860
35. Partin AW, Criley SR, Subong EN et al: Standard versus age-specific prostate specific antigen reference ranges among men with clinically localized prostate cancer: A pathological analysis. J Urol 1996; **155:** 1336
36. Guess HA, Gormley GJ, Stoner E and Oesterling JE: The effect of finasteride on prostate specific antigen: Review of available data. J Urol 1996; **155:** 3
37. Brawer MK, Lin DW, Williford WO et al: Effect of finasteride and/or terazosin on serum psa: Results of va cooperative study #359. Prostate 1999; **39:** 234
38. Franlund M, Mansson M, Godtman RA et al: Results from 22 years of followup in the goteborg randomized population-based prostate cancer screening trial. J Urol 2022; **208:** 292
39. Loeb S, Carter HB, Catalona WJ et al: Baseline prostate-specific antigen testing at a young age. Eur Urol 2012; **61:** 1
40. Vickers AJ, Ulmert D, Sjoberg DD et al: Strategy for detection of prostate cancer based on relation between prostate specific antigen at age 40-55 and long term risk of metastasis: Case-control study. BMJ 2013; **346:** f2023
41. Arsov C, Albers P, Herkommer K et al: A randomized trial of risk-adapted screening for prostate cancer in young men-results of the first screening round of the probase trial. Int J Cancer 2022; **150:** 1861
42. Gulati R, Cheng HH, Lange PH et al: Screening men at increased risk for prostate cancer diagnosis: Model estimates of benefits and harms. Cancer Epidemiology, Biomarkers & Prevention 2017; **26:** 222
43. Giaquinto AN, Miller KD, Tossas KY et al: Cancer statistics for african american/black people 2022. CA Cancer J Clin 2022; **72:** 202
44. Tsodikov A, Gulati R, de Carvalho TM et al: Is prostate cancer different in black men? Answers from 3 natural history models. Cancer 2017; **123:** 2312
45. Bancroft EK, Page EC, Brook MN et al: A prospective prostate cancer screening programme for men with pathogenic variants in mismatch repair genes (impact): Initial results from an international prospective study. Lancet Oncol 2021; **22:** 1618
46. Mitra AV, Bancroft EK, Barbachano Y et al: Targeted prostate cancer screening in men with mutations in brca1 and brca2 detects aggressive prostate cancer: Preliminary analysis of the results of the impact study. BJU International 2011; **107:** 28
47. Page EC, Bancroft EK, Brook MN et al: Interim results from the impact study: Evidence for prostate-specific antigen screening in brca2 mutation carriers. European Urology 2019; **76:** 831
48. Schaeffer EM, Srinivas S, Adra N et al: Nccn guidelines® insights: Prostate cancer, version 1.2023. J Natl Compr Canc Netw 2022; **20:** 1288
49. Giri VN, Knudsen KE, Kelly WK et al: Role of genetic testing for inherited prostate cancer risk: Philadelphia prostate cancer consensus conference 2017. J Clin Oncol 2018; **36:** 414
50. Albright F, Stephenson RA, Agarwal N et al: Prostate cancer risk prediction based on complete prostate cancer family history. Prostate 2015; **75:** 390
51. Tangen CM, Goodman PJ, Till C et al: Biases in recommendations for and acceptance of prostate biopsy significantly affect assessment of prostate cancer risk factors: Results from two large randomized clinical trials. J Clin Oncol 2016; **34:** 4338
52. Clements MB, Vertosick EA, Guerrios-Rivera L et al: Defining the impact of family history on detection of high-grade prostate cancer in a large multi-institutional cohort. Eur Urol 2022; **82:** 163
53. Bratt O, Drevin L, Akre O et al: Family history and probability of prostate cancer, differentiated by risk category: A nationwide population-based study. J Natl Cancer Inst 2016; **108:** djw110
54. Barber L, Gerke T, Markt SC et al: Family history of breast or prostate cancer and prostate cancer risk. Clin Cancer Res 2018; **24:** 5910

55. Thomas JA, 2nd, Gerber L, Moreira DM et al: Prostate cancer risk in men with prostate and breast cancer family history: Results from the reduce study (r1). J Intern Med 2012; **272:** 85

56. Pritchard CC, Mateo J, Walsh MF et al: Inherited DNA-repair gene mutations in men with metastatic prostate cancer. N Engl J Med 2016; **375:** 443

57. Hugosson J, Carlsson S, Aus G et al: Mortality results from the goteborg randomised population-based prostate-cancer screening trial. Lancet Oncol 2010; **11:** 725

58. Schroder FH, Hugosson J, Roobol MJ et al: Screening and prostate-cancer mortality in a randomized european study. New England Journal of Medicine 2009; **360:** 1320

59. Heijnsdijk EA, Wever EM, Auvinen A et al: Quality-of-life effects of prostate-specific antigen screening. N Engl J Med 2012; **367:** 595

60. Carlsson S, Assel M, Sjoberg D et al: Influence of blood prostate specific antigen levels at age 60 on benefits and harms of prostate cancer screening: Population based cohort study. BMJ 2014; **348:** g2296

61. Pinsky PF, Miller E, Prorok P et al: Extended follow-up for prostate cancer incidence and mortality among participants in the prostate, lung, colorectal and ovarian randomized cancer screening trial. BJU Int 2019; **123:** 854

62. Shoag JE, Mittal S and Hu JC: Reevaluating psa testing rates in the plco trial. N Engl J Med 2016; **374:** 1795

63. Tsodikov A, Gulati R, Heijnsdijk EAM et al: Reconciling the effects of screening on prostate cancer mortality in the erspc and plco trials. Ann Intern Med 2017; **167:** 449

64. Roobol MJ, Roobol DW and Schroder FH: Is additional testing necessary in men with prostate-specific antigen levels of 1.0 ng/ml or less in a population-based screening setting? (erspc, section rotterdam). Urology 2005; **65:** 343

65. Vickers AJ, Cronin AM, Bjork T et al: Prostate specific antigen concentration at age 60 and death or metastasis from prostate cancer: Case-control study. BMJ 2010; **341:** c4521

66. Vertosick EA, Haggstrom C, Sjoberg DD et al: Prespecified 4-kallikrein marker model at age 50 or 60 for early detection of lethal prostate cancer in a large population based cohort of asymptomatic men followed for 20 years. J Urol 2020; **204:** 281

67. Preston MA, Gerke T, Carlsson SV et al: Baseline prostate-specific antigen level in midlife and aggressive prostate cancer in black men. Eur Urol 2019; **75:** 399

68. Preston MA, Batista JL, Wilson KM et al: Baseline prostate-specific antigen levels in midlife predict lethal prostate cancer. J Clin Oncol 2016; **34:** 2705

69. Kovac E, Carlsson SV, Lilja H et al: Association of baseline prostate-specific antigen level with long-term diagnosis of clinically significant prostate cancer among patients aged 55 to 60 years: A secondary analysis of a cohort in the prostate, lung, colorectal, and ovarian (plco) cancer screening trial. JAMA Netw Open 2020; **3:** e1919284

70. Ross KS, Carter HB, Pearson JD and Guess HA: Comparative efficiency of prostate-specific antigen screening strategies for prostate cancer detection. JAMA 2000; **284:** 1399

71. Heijnsdijk EAM, Gulati R, Tsodikov A et al: Lifetime benefits and harms of prostate-specific antigen-based risk-stratified screening for prostate cancer. Journal of the National Cancer Institute 2020; **112:** 1013

72. Remmers S, Bangma CH, Godtman RA et al: Relationship between baseline prostate-specific antigen on cancer detection and prostate cancer death: Long-term follow-up from the european randomized study of screening for prostate cancer. Eur Urol 2023; **84:** 503

73. Grenabo Bergdahl A, Holmberg E, Moss S and Hugosson J: Incidence of prostate cancer after termination of screening in a population-based randomised screening trial. European Urology 2013; **64:** 703

74. Godtman RA, Kollberg KS, Pihl CG et al: The association between age, prostate cancer risk, and higher gleason score in a long-term screening program: Results from the goteborg-1 prostate cancer screening trial. Eur Urol 2022; **82:** 311

75. Schaeffer EM, Carter HB, Kettermann A et al: Prostate specific antigen testing among the elderly--when to stop? Journal of Urology 2009; **181:** 1606

76. de V, II, Remmers S, Hogenhout R and Roobol MJ: Prostate cancer mortality among elderly men after discontinuing organised screening: Long-term results from the european randomized study of screening for prostate cancer rotterdam. Eur Urol 2024; **85:** 74

77. Lansdorp-Vogelaar I, Gulati R, Mariotto AB et al: Personalizing age of cancer screening cessation based on comorbid conditions: Model estimates of harms and benefits. Ann Intern Med 2014; **161:** 104

78. Schroder FH, Hugosson J, Roobol MJ et al: Prostate-cancer mortality at 11 years of follow-up. New England Journal of Medicine 2012; **366:** 981

79. Bill-Axelson A, Holmberg L, Garmo H et al: Radical prostatectomy or watchful waiting in early prostate cancer. N Engl J Med 2014; **370:** 932

80. Vickers A, Bennette C, Steineck G et al: Individualized estimation of the benefit of radical prostatectomy from the scandinavian prostate cancer group randomized trial. Eur Urol 2012; **62:** 204

81. Hamdy FC, Donovan JL, Lane JA et al: 10-year outcomes after monitoring, surgery, or radiotherapy for localized prostate cancer. N Engl J Med 2016; **375:** 1415

82. Wilt TJ, Vo TN, Langsetmo L et al: Radical prostatectomy or observation for clinically localized prostate cancer: Extended follow-up of the prostate cancer intervention versus observation trial (pivot). Eur Urol 2020; **77:** 713

83. Sakr WA, Grignon DJ, Crissman JD et al: High grade prostatic intraepithelial neoplasia (hgpin) and prostatic adenocarcinoma between the ages of 20-69: An autopsy study of 249 cases. In Vivo 1994; **8:** 439

84. Vickers AJ, Sjoberg DD, Ulmert D et al: Empirical estimates of prostate cancer overdiagnosis by age and prostate-specific antigen. BMC Med 2014; **12:** 26

85. Gulati R, Feuer EJ and Etzioni R: Conditions for valid empirical estimates of cancer overdiagnosis in randomized trials and population studies. Am J Epidemiol 2016; **184:** 140

86. Wilson JR, Clarke MG, Ewings P et al: The assessment of patient life-expectancy: How accurate are urologists and oncologists? BJU Int 2005; **95:** 794

87. Makarov D, A F, J F et al: Implementation of shared decision making into urological practice, 2022

88. Matsukawa A, Yanagisawa T, Bekku K et al: Comparing the performance of digital rectal examination and prostate-specific antigen as a screening test for prostate cancer: A systematic review and meta-analysis. Eur Urol Oncol 2024; **7:** 697

89. Thompson IM, Tangen CM, Goodman PJ et al: Finasteride improves the sensitivity of digital rectal examination for prostate cancer detection. J Urol 2007; **177:** 1749

90. Nagler HM, Gerber EW, Homel P et al: Digital rectal examination is barrier to population-based prostate cancer screening. Urology 2005; **65:** 1137

91. Gosselaar C, Roobol MJ, Roemeling S and Schroder FH: The role of the digital rectal examination in subsequent screening visits in the european randomized study of screening for prostate cancer (erspc), rotterdam. European Urology 2008; **54:** 581

92. Luboldt HJ, Bex A, Swoboda A et al: Early detection of prostate cancer in germany: A study using digital rectal examination and 4.0 ng/ml prostate-specific antigen as cutoff. European Urology 2001; **39:** 131

93. Halpern JA, Oromendia C, Shoag JE et al: Use of digital rectal examination as an adjunct to prostate specific antigen in the detection of clinically significant prostate cancer. J Urol 2018; **199:** 947

94. Vickers AJ, Till C, Tangen CM et al: An empirical evaluation of guidelines on prostate-specific antigen velocity in prostate cancer detection. Journal of the National Cancer Institute 2011; **103:** 462

95. Vickers AJ, Wolters T, Savage CJ et al: Prostate-specific antigen velocity for early detection of prostate cancer: Result from a large, representative, population-based cohort. European Urology 2009; **56:** 753

96. Eggener SE, Yossepowitch O, Roehl KA et al: Relationship of prostate-specific antigen velocity to histologic findin a prostate cancer screening program. Urology 2008; **71:** 1016

97. Carbunaru S, Nettey OS, Gogana P et al: A comparative effectiveness analysis of the pbcg vs. Pcpt risks calculators in a multi-ethnic cohort. BMC Urol 2019; **19:** 121

98. Breza J, Subin F, Bernadic M et al: The use of european randomized study of screening for prostate cancer calculator as a diagnostic tool for prostate biopsy indication. Bratisl Lek Listy 2019; **120:** 331

99. Thompson IM, Ankerst DP, Chi C et al: Assessing prostate cancer risk: Results from the prostate cancer prevention trial. J Natl Cancer Inst 2006; **98:** 529

100. Ankerst DP, Goros M, Tomlins SA et al: Incorporation of urinary prostate cancer antigen 3 and tmprss2:Erg into prostate cancer prevention trial risk calculator. Eur Urol Focus 2019; **5:** 54

101. Chun FK, Briganti A, Graefen M et al: Development and external validation of an extended 10-core biopsy nomogram. Eur Urol 2007; **52:** 436

102. Perdonà S, Cavadas V, Di Lorenzo G et al: Prostate cancer detection in the "grey area" of prostate-specific antigen below 10 ng/ml: Head-to-head comparison of the updated pcpt calculator and chun's nomogram, two risk estimators incorporating prostate cancer antigen 3. Eur Urol 2011; **59:** 81

103. Kranse R, Roobol M and Schröder FH: A graphical device to represent the outcomes of a logistic regression analysis. Prostate 2008; **68:** 1674

104. Roobol MJ, Schröder FH, Hugosson J et al: Importance of prostate volume in the european randomised study of screening for prostate cancer (erspc) risk calculators: Results from the prostate biopsy collaborative group. World J Urol 2012; **30:** 149

105. De Nunzio C, Lombardo R, Tema G et al: External validation of chun, pcpt, erspc, kawakami, and karakiewicz nomograms in the prediction of prostate cancer: A single center cohort-study. Urol Oncol 2018; **36:** 364.e1

106. Ankerst DP, Straubinger J, Selig K et al: A contemporary prostate biopsy risk calculator based on multiple heterogeneous cohorts. Eur Urol 2018; **74:** 197

107. Ankerst DP, Hoefler J, Bock S et al: Prostate cancer prevention trial risk calculator 2.0 for the prediction of low- vs high-grade prostate cancer. Urology 2014; **83:** 1362

108. Fenton JJ, Weyrich MS, Durbin S et al: U.S. Preventive services task force evidence syntheses, formerly systematic evidence reviews. In: Prostate-specific antigen-based screening for prostate cancer: A systematic evidence review for the u.S. Preventive services task force. Rockville (MD): Agency for Healthcare Research and Quality (US), 2018

109. Wei JT, Feng Z, Partin AW et al: Can urinary pca3 supplement psa in the early detection of prostate cancer? Journal of Clinical Oncology 2014; **32:** 4066

110. Singhal U, Tosoian JJ, Qi J et al: Overtreatment and underutilization of watchful waiting in men with limited life expectancy: An analysis of the michigan urological surgery improvement collaborative registry. Urology 2020; **145:** 190

111. Kasivisvanathan V, Rannikko AS, Borghi M et al: Mri-targeted or standard biopsy for prostate-cancer diagnosis. N Engl J Med 2018; **378:** 1767

112. Wei C, Szewczyk-Bieda M, Bates AS et al: Multicenter randomized trial assessing mri and image-guided biopsy for suspected prostate cancer: The multipros study. Radiology 2023; **308:** e221428

113. Hugosson J, Godtman RA, Wallstrom J et al: Results after four years of screening for prostate cancer with psa and mri. N Engl J Med 2024; **391:** 1083

114. Hamm CA, Asbach P, Pöhlmann A et al: Oncological safety of mri-informed biopsy decision-making in men with suspected prostate cancer. JAMA Oncol 2025; **11:** 145

115. Boschheidgen M, Albers P, Schlemmer HP et al: Multiparametric magnetic resonance imaging in prostate cancer screening at the age of 45 years: Results from the first screening round of the probase trial. Eur Urol 2024; **85:** 105

116. Panebianco V, Barchetti F, Sciarra A et al: Multiparametric magnetic resonance imaging vs. Standard care in men being evaluated for prostate cancer: A randomized study. Urol Oncol 2015; **33:** 17 e1

117. Tonttila PP, Lantto J, Pääkkö E et al: Prebiopsy multiparametric magnetic resonance imaging for prostate cancer diagnosis in biopsy-naive men with suspected prostate cancer based on elevated prostate-specific antigen values: Results from a randomized prospective blinded controlled trial. Eur Urol 2016; **69:** 419

118. Porpiglia F, Manfredi M, Mele F et al: Diagnostic pathway with multiparametric magnetic resonance imaging versus standard pathway: Results from a randomized prospective study in biopsy-naive patients with suspected prostate cancer. Eur Urol 2017; **72:** 282

119. Hugosson J, Månsson M, Wallström J et al: Prostate cancer screening with psa and mri followed by targeted biopsy only. N Engl J Med 2022; **387:** 2126

120. Drost FH, Osses DF, Nieboer D et al: Prostate mri, with or without mri-targeted biopsy, and systematic biopsy for detecting prostate cancer. Cochrane Database of Systematic Reviews 2019; **4:** CD012663

121. Purysko AS, Tempany C, Macura KJ et al: American college of radiology initiatives on prostate magnetic resonance imaging quality. Eur J Radiol 2023; **165:** 110937

122. Barentsz JO, Richenberg J, Clements R et al: Esur prostate mr guidelines 2012. Eur Radiol 2012; **22:** 746

123. Weinreb JC, Barentsz JO, Choyke PL et al: Pi-rads prostate imaging - reporting and data system: 2015, version 2. Eur Urol 2016; **69:** 16

124. Turkbey B, Rosenkrantz AB, Haider MA et al: Prostate imaging reporting and data system version 2.1: 2019 update of prostate imaging reporting and data system version 2. Eur Urol 2019; **76:** 340

125. Ahdoot M, Lebastchi AH, Long L et al: Using prostate imaging-reporting and data system (pi-rads) scores to select an optimal prostate biopsy method: A secondary analysis of the trio study. European Urology Oncology 2021; **10:** 10

126. Barkovich EJ, Shankar PR and Westphalen AC: A systematic review of the existing prostate imaging reporting and data system version 2 (pi-radsv2) literature and subset meta-analysis of pi-radsv2 categories stratified by gleason scores. AJR American Journal of Roentgenology 2019; **212:** 847

127. Benndorf M, Waibel L, Kronig M et al: Peripheral zone lesions of intermediary risk in multiparametric prostate mri: Frequency and validation of the pi-radsv2 risk stratification algorithm based on focal contrast enhancement. European Journal of Radiology 2018; **99:** 62

128. Borkowetz A, Hadaschik B, Platzek I et al: Prospective comparison of transperineal magnetic resonance imaging/ultrasonography fusion biopsy and transrectal systematic biopsy in biopsy-naive patients. BJU International 2018; **121:** 53

129. Chen Y, Ruan M, Zhou B et al: Cutoff values of prostate imaging reporting and data system version 2.1 score in men with prostate-specific antigen level 4 to 10 ng/ml: Importance of lesion location. Clinical Genitourinary Cancer 2021; **19:** 288

130. Cheng Y, Qi F, Liang L et al: Use of prostate systematic and targeted biopsy on the basis of bi-parametric magnetic resonance imaging in biopsy-naive patients. Journal of Investigative Surgery 2020; **35:** 92

131. Daun M, Fardin S, Ushinsky A et al: Pi-rads version 2 is an excellent screening tool for clinically significant prostate cancer as designated by the validated international society of urological pathology criteria: A retrospective analysis. Current Problems in Diagnostic Radiology 2020; **49:** 407

132. Hofbauer SL, Maxeiner A, Kittner B et al: Validation of prostate imaging reporting and data system version 2 for the detection of prostate cancer. Journal of Urology 2018; **200:** 767

133. John S, Cooper S, Breau RH et al: Multiparametric magnetic resonance imaging-transrectal ultrasound-guided cognitive fusion biopsy of the prostate: Clinically significant cancer detection rates stratified by the prostate imaging and data reporting system version 2 assessment categories. Canadian Urological Association Journal 2018; **12:** 401

134. Jordan EJ, Fiske C, Zagoria RJ and Westphalen AC: Evaluating the performance of pi-rads v2 in the non-academic setting. Abdominal Radiology 2017; **42:** 2725

135. Kim M, Ryu H, Lee HJ et al: Who can safely evade a magnetic resonance imaging fusion-targeted biopsy (mriftb) for prostate imaging reporting and data system (pi-rads) 3 lesion? World Journal of Urology 2021; **39:** 1463

136. Lee CU, Choi J, Sung SH et al: The role of prostate combination biopsy consisting of targeted and additional systematic biopsy. Journal of Clinical Medicine 2021; **10:** 4804

137. Lim C, Abreu-Gomez J, Leblond MA et al: When to biopsy prostate imaging and data reporting system version 2 (pi-radsv2) assessment category 3 lesions? Use of clinical and imaging variables to predict cancer diagnosis at targeted biopsy. Canadian Urological Association Journal 2020; **15:** 115

138. Mathur S, O'Malley ME, Ghai S et al: Correlation of 3t multiparametric prostate mri using prostate imaging reporting and data system (pirads) version 2 with biopsy as reference standard. Abdominal Radiology 2019; **44:** 252

139. Osses DF, van Asten JJ, Kieft GJ and Tijsterman JD: Prostate cancer detection rates of magnetic resonance imaging-guided prostate biopsy related to prostate imaging reporting and data system score. World Journal of Urology 2017; **35:** 207

140. Park BK and Park SY: New biopsy techniques and imaging features of transrectal ultrasound for targeting pi-rads 4 and 5 lesions. Journal of Clinical Medicine 2020; **9:** 15

141. Ryoo H, Kang MY, Sung HH et al: Detection of prostate cancer using prostate imaging reporting and data system score and prostate-specific antigen density in biopsy-naive and prior biopsy-negative patients. Prostate International 2020; **8:** 125

142. Sathianathen NJ, Konety BR, Soubra A et al: Which scores need a core? An evaluation of mr-targeted biopsy yield by pirads score across different biopsy indications. Prostate Cancer & Prostatic Diseases 2018; **21:** 573

143. Sokhi HK, Padhani AR, Patel S and Pope A: Diagnostic yields in patients with suspected prostate cancer undergoing mri as the first-line investigation in routine practice. Clinical Radiology 2020; **75:** 950

144. Syed JS, Nguyen KA, Nawaf CB et al: Prostate zonal anatomy correlates with the detection of prostate cancer on multiparametric magnetic resonance imaging/ultrasound fusion-targeted biopsy in patients with a solitary pi-rads v2-scored lesion. Urologic Oncology 2017; **35:** 542.e19

145. Westphalen AC, McCulloch CE, Anaokar JM et al: Variability of the positive predictive value of pi-rads for prostate mri across 26 centers: Experience of the society of abdominal radiology prostate cancer disease-focused panel. Radiology 2020; **296:** 76

146. Abdul Raheem R, Razzaq A, Beraud V et al: Can a prostate biopsy be safely deferred on pi-rads 1,2 or 3 lesions seen on pre-biopsy mp-mri? Arab Journal of Urology 2022; **21:** 10

147. Nowier A, Mazhar H, Salah R and Shabayek M: Performance of multi-parametric magnetic resonance imaging through pirads scoring system in biopsy naive patients with suspicious prostate cancer. Arab Journal of Urology Print 2022; **20:** 121

148. Rosenkrantz AB, Ayoola A, Hoffman D et al: The learning curve in prostate mri interpretation: Self-directed learning versus continual reader feedback. AJR Am J Roentgenol 2017; **208:** W92

149. Wei CG, Zhang YY, Pan P et al: Diagnostic accuracy and interobserver agreement of pi-rads version 2 and version 2.1 for the detection of transition zone prostate cancers. AJR Am J Roentgenol 2021; **216:** 1247

150. Bhayana R, O'Shea A, Anderson MA et al: Pi-rads versions 2 and 2.1: Interobserver agreement and diagnostic performance in peripheral and transition zone lesions among six radiologists. AJR Am J Roentgenol 2021; **217:** 141

151. Annamalai A, Fustok JN, Beltran-Perez J et al: Interobserver agreement and accuracy in interpreting mpmri of the prostate: A systematic review. Curr Urol Rep 2022; **23:** 1

152. de Rooij M, Israel B, Tummers M et al: Esur/esui consensus statements on multi-parametric mri for the detection of clinically significant prostate cancer: Quality requirements for image acquisition, interpretation and radiologists' training. Eur Radiol 2020; **30:** 5404

153. Li JL, Phillips D, Towfighi S et al: Second-opinion reads in prostate mri: Added value of subspecialty interpretation and review at multidisciplinary rounds. Abdom Radiol (NY) 2022; **47:** 827

154. Al Hussein Al Awamlh B, Marks LS, Sonn GA et al: Multicenter analysis of clinical and mri characteristics associated with detecting clinically significant prostate cancer in pi-rads (v2.0) category 3 lesions. Urologic Oncology 2020; **38:** 637.e9

155. Cata E, Andras I, Ferro M et al: Systematic sampling during mri-us fusion prostate biopsy can overcome errors of targeting-prospective single center experience after 300 cases in first biopsy setting. Translational Andrology & Urology 2020; **9:** 2510

156. Filson CP, Natarajan S, Margolis DJ et al: Prostate cancer detection with magnetic resonance-ultrasound fusion bic The role of systematic and targeted biopsies. Cancer 2016; **122:** 884

157. Fujii S, Hayashi T, Honda Y et al: Magnetic resonance imaging/transrectal ultrasonography fusion targeted prostate biopsy finds more significant prostate cancer in biopsy-naive japanese men compared with the standard biopsy.

International Journal of Urology 2020; **27:** 140

158. Kim YJ, Huh JS and Park KK: Effectiveness of bi-parametric mr/us fusion biopsy for detecting clinically significant prostate cancer in prostate biopsy naive men. Yonsei Medical Journal 2019; **60:** 346

159. Maxeiner A, Kittner B, Blobel C et al: Primary magnetic resonance imaging/ultrasonography fusion-guided biopsy of the prostate. BJU International 2018; **122:** 211

160. Ahdoot M, Wilbur AR, Reese SE et al: Mri-targeted, systematic, and combined biopsy for prostate cancer diagnosis. New England Journal of Medicine 2020; **382:** 917

161. Coker MA, Glaser ZA, Gordetsky JB et al: Targets missed: Predictors of mri-targeted biopsy failing to accurately localize prostate cancer found on systematic biopsy. Prostate Cancer Prostatic Dis 2018; **21:** 549

162. Brisbane WG, Priester AM, Ballon J et al: Targeted prostate biopsy: Umbra, penumbra, and value of perilesional sampling. Eur Urol 2022; **82:** 303

163. Williams C, Ahdoot M, Daneshvar MA et al: Why does magnetic resonance imaging-targeted biopsy miss clinically significant cancer? J Urol 2022; **207:** 95

164. Prince M, Foster BR, Kaempf A et al: In-bore versus fusion mri-targeted prostate biopsy of pi-rads category 4 or 5 lesions: A retrospective comparative analysis using propensity score weighting. AJR Am J Roentgenol 2021; **217:** 1123

165. Sathianathen NJ, Omer A, Harriss E et al: Negative predictive value of multiparametric magnetic resonance imaging in the detection of clinically significant prostate cancer in the prostate imaging reporting and data system era: A systematic review and meta-analysis. European Urology 2020; **78:** 402

166. Haj-Mirzaian A, Burk KS, Lacson R et al: Magnetic resonance imaging, clinical, and biopsy findings in suspected prostate cancer: A systematic review and meta-analysis. JAMA Netw Open 2024; **7:** e244258

167. Eichler K, Hempel S, Wilby J et al: Diagnostic value of systematic biopsy methods in the investigation of prostate cancer: A systematic review. J Urol 2006; **175:** 1605

168. Ristau BT, Allaway M, Cendo D et al: Free-hand transperineal prostate biopsy provides acceptable cancer detection and minimizes risk of infection: Evolving experience with a 10-sector template. Urol Oncol 2018; **36:** 528.e15

169. Ukimura O, Coleman JA, de la Taille A et al: Contemporary role of systematic prostate biopsies: Indications, techniques, and implications for patient care. Eur Urol 2013; **63:** 214

170. Presti JC, Jr., O'Dowd GJ, Miller MC et al: Extended peripheral zone biopsy schemes increase cancer detection rates and minimize variance in prostate specific antigen and age related cancer rates: Results of a community multi-practice study. J Urol 2003; **169:** 125

171. Catalona WJ, Partin AW, Slawin KM et al: Use of the percentage of free prostate-specific antigen to enhance differentiation of prostate cancer from benign prostatic disease: A prospective multicenter clinical trial. JAMA 1998; **279:** 1542

172. Roddam AW, Duffy MJ, Hamdy FC et al: Use of prostate-specific antigen (psa) isoforms for the detection of prostate cancer in men with a psa level of 2-10 ng/ml: Systematic review and meta-analysis. Eur Urol 2005; **48:** 386

173. Lee R, Localio AR, Armstrong K et al: A meta-analysis of the performance characteristics of the free prostate-specific antigen test. Urology 2006; **67:** 762

174. Catalona WJ, Southwick PC, Slawin KM et al: Comparison of percent free psa, psa density, and age-specific psa cutoffs for prostate cancer detection and staging. Urology 2000; **56:** 255

175. Bruzzese D, Mazzarella C, Ferro M et al: Prostate health index vs percent free prostate-specific antigen for prostate cancer detection in men with "gray" prostate-specific antigen levels at first biopsy: Systematic review and meta-analysis. Transl Res 2014; **164:** 444

176. Finne P, Auvinen A, Maattanen L et al: Diagnostic value of free prostate-specific antigen among men with a prostate-specific antigen level of <3.0 microg per liter. European Urology 2008; **54:** 362

177. Parekh DJ, Punnen S, Sjoberg DD et al: A multi-institutional prospective trial in the USA confirms that the 4kscore accurately identifies men with high-grade prostate cancer. European Urology 2015; **68:** 464

178. Loeb S, Sanda MG, Broyles DL et al: The prostate health index selectively identifies clinically significant prostate cancer. Journal of Urology 2015; **193:** 1163

179. Morote J, Celma A, Planas J et al: Diagnostic accuracy of prostate health index to identify aggressive prostate cancer. An institutional validation study. Actas Urologicas Espanolas 2016; **40:** 378

180. Seisen T, Roupret M, Brault D et al: Accuracy of the prostate health index versus the urinary prostate cancer antigen 3 score to predict overall and significant prostate cancer at initial biopsy. Prostate 2015; **75:** 103

181. Kim L, Boxall N, George A et al: Clinical utility and cost modelling of the phi test to triage referrals into image-based diagnostic services for suspected prostate cancer: The prim (phi to refine mri) study. BMC Medicine 2020; **18:** 95

182. Tosoian JJ, Zhang Y, Xiao L et al: Development and validation of an 18-gene urine test for high-grade prostate cancer. JAMA Oncology 2024; **10:** 726

183. Eyrich NW, Morgan TM and Tosoian JJ: Biomarkers for detection of clinically significant prostate cancer: Contemporary clinical data and future directions. Transl Androl Urol 2021; **10:** 3091

184. de la Calle C, Patil D, Wei JT et al: Multicenter evaluation of the prostate health index to detect aggressive prostate cancer in biopsy naive men. Journal of Urology 2015; **194:** 65

185. de la Taille A, Irani J, Graefen M et al: Clinical evaluation of the pca3 assay in guiding initial biopsy decisions. Journal of Urology 2011; **185:** 2119

186. Falagario UG, Martini A, Wajswol E et al: Avoiding unnecessary magnetic resonance imaging (mri) and biopsies: Negative and positive predictive value of mri according to prostate-specific antigen density, 4kscore and risk calculators. European Urology Oncology 2020; **3:** 700

187. Hansen J, Auprich M, Ahyai SA et al: Initial prostate biopsy: Development and internal validation of a biopsy-specific nomogram based on the prostate cancer antigen 3 assay. European Urology 2013; **63:** 201

188. Hendriks RJ, van der Leest MMG, Israel B et al: Clinical use of the selectmdx urinary-biomarker test with or without mpmri in prostate cancer diagnosis: A prospective, multicenter study in biopsy-naive men. Prostate Cancer & Prostatic Diseases 2021; **3:** 03

189. Lazzeri M, Haese A, de la Taille A et al: Serum isoform [-2]propsa derivatives significantly improve prediction of prostate cancer at initial biopsy in a total psa range of 2-10 ng/ml: A multicentric european study. Eur Urol 2013; **63:** 986

190. Lebastchi AH, Russell CM, Niknafs YS et al: Impact of the myprostatescore (mps) test on the clinical decision to undergo prostate biopsy: Results from a contemporary academic practice. Urology 2020; **145:** 204

191. Lendinez-Cano G, Ojeda-Claro AV, Gomez-Gomez E et al: Prospective study of diagnostic accuracy in the detection of high-grade prostate cancer in biopsy-naive patients with clinical suspicion of prostate cancer who underwent the select mdx test. Prostate 2021; **81:** 857

192. McKiernan J, Donovan MJ, Margolis E et al: A prospective adaptive utility trial to validate performance of a novel urine exosome gene expression assay to predict high-grade prostate cancer in patients with prostate-specific antigen 2-10ng/ml at initial biopsy. European Urology 2018; **74:** 731

193. McKiernan J, Donovan MJ, O'Neill V et al: A novel urine exosome gene expression assay to predict high-grade prostate cancer at initial biopsy. JAMA Oncology 2016; **2:** 882

194. Nordstrom T, Vickers A, Assel M et al: Comparison between the four-kallikrein panel and prostate health index for predicting prostate cancer. European Urology 2015; **68:** 139

195. Tutrone R, Donovan MJ, Torkler P et al: Clinical utility of the exosome based exodx prostate(intelliscore) epi test in men presenting for initial biopsy with a psa 2-10 ng/ml. Prostate Cancer & Prostatic Diseases 2020; **23:** 607

196. Rubio-Briones J, Borque A, Esteban LM et al: Optimizing the clinical utility of pca3 to diagnose prostate cancer in initial prostate biopsy. BMC Cancer 2015; **15:** 633

197. Jiao B, Gulati R, Hendrix N et al: Economic evaluation of urine-based or magnetic resonance imaging reflex tests in men with intermediate prostate-specific antigen levels in the united states. Value Health 2021; **24:** 1111

198. Benchikh A, Savage C, Cronin A et al: A panel of kallikrein markers can predict outcome of prostate biopsy following clinical work-up: An independent validation study from the european randomized study of prostate cancer screening,

france. BMC Cancer 2010; **10:** 635

199. Gupta A, Roobol MJ, Savage CJ et al: A four-kallikrein panel for the prediction of repeat prostate biopsy: Data from the european randomized study of prostate cancer screening in rotterdam, netherlands. British Journal of Cancer 2010; **103:** 708

200. Josefsson A, Mansson M, Kohestani K et al: Performance of 4kscore as a reflex test to prostate-specific antigen in the goteborg-2 prostate cancer screening trial. European Urology 2024; **86:** 223

201. de Almeida SR, Jr., Thomas J, Mason MM et al: Optimum threshold of the 4kscore for biopsy in men with negative or indeterminate multiparametric magnetic resonance imaging. BJUI Compass 2023; **4:** 591

202. Thomas J, Atluri S, Zucker I et al: A multi-institutional study of 1,111 men with 4k score, multiparametric magnetic resonance imaging, and prostate biopsy. Urologic Oncology 2023; **41:** 430.e9

203. Stovsky M, Klein EA, Chait A et al: Clinical validation of isopsa™, a single parameter, structure based assay for improved detection of high grade prostate cancer. J Urol 2019; **201:** 1115

204. Benidir T, Lone Z, Wood A et al: Using isopsa with prostate imaging reporting and data system score may help refine biopsy decision making in patients with elevated psa. Urology 2023; **176:** 115

205. Klein EA, Chait A, Hafron JM et al: The single-parameter, structure-based isopsa assay demonstrates improved diagnostic accuracy for detection of any prostate cancer and high-grade prostate cancer compared to a concentration-based assay of total prostate-specific antigen: A preliminary report. Eur Urol 2017; **72:** 942

206. Klein EA, Partin A, Lotan Y et al: Clinical validation of isopsa, a single parameter, structure-focused assay for improved detection of prostate cancer: A prospective, multicenter study. Urol Oncol 2022; **40:** 408.e9

207. Steuber T, Heidegger I, Kafka M et al: Propose: A real-life prospective study of proclarix, a novel blood-based test to support challenging biopsy decision-making in prostate cancer. Eur Urol Oncol 2022; **5:** 321

208. Kaufmann B, Fischer S, Athanasiou A et al: Evaluation of proclarix in the diagnostic work-up of prostate cancer. BJUI Compass 2024; **5:** 297

209. Morote J, Pye H, Campistol M et al: Accurate diagnosis of prostate cancer by combining proclarix with magnetic resonance imaging. BJU International 2023; **132:** 188

210. Lazzeri M, Briganti A, Scattoni V et al: Serum index test %[-2]propsa and prostate health index are more accurate than prostate specific antigen and %fpsa in predicting a positive repeat prostate biopsy. Journal of Urology 2012; **188:** 1137

211. Gnanapragasam VJ, Burling K, George A et al: The prostate health index adds predictive value to multi-parametric mri in detecting significant prostate cancers in a repeat biopsy population. Scientific Reports 2016; **6:** 35364

212. Porpiglia F, Russo F, Manfredi M et al: The roles of multiparametric magnetic resonance imaging, pca3 and prostate health index-which is the best predictor of prostate cancer after a negative biopsy? Journal of Urology 2014; **192:** 60

213. Zhou Y, Fu Q, Shao Z et al: The function of prostate health index in detecting clinically significant prostate cancer in the pi-rads 3 population: A multicenter prospective study. World Journal of Urology 2023; **41:** 455

214. Zhu M, Fu Q, Zang Y et al: Different diagnostic strategies combining prostate health index and magnetic resonance imaging for predicting prostate cancer: A multicentre study. Urologic Oncology 2024; **42:** 159.e17

215. Chiu PK, Lam TY, Ng CF et al: The combined role of mri prostate and prostate health index in improving detection of significant prostate cancer in a screening population of chinese men. Asian Journal of Andrology 2023; **25:** 674

216. Boo Y, Chung JH, Kang M et al: Comparison of prostate-specific antigen and its density and prostate health index and its density for detection of prostate cancer. Biomedicines 2023; **11:** 06

217. Chiu PK, Leow JJ, Chiang CH et al: Prostate health index density outperforms prostate-specific antigen density in the diagnosis of clinically significant prostate cancer in equivocal magnetic resonance imaging of the prostate: A multicenter evaluation. Journal of Urology 2023; **210:** 88

218. Chiu PK, Liu AQ, Lau SY et al: A 2-year prospective evaluation of the prostate health index in guiding biopsy decis in a large cohort. BJU International 2024; **4:** 04

219. Strom P, Nordstrom T, Aly M et al: The stockholm-3 model for prostate cancer detection: Algorithm update, biomarker contribution, and reflex test potential. European Urology 2018; **74:** 204

220. Bjornebo L, Discacciati A, Falagario U et al: Biomarker vs mri-enhanced strategies for prostate cancer screening: The sthlm3-mri randomized clinical trial. JAMA Network Open 2024; **7:** e247131

221. Elyan A, Saba K, Sigle A et al: Prospective multicenter validation of the stockholm3 test in a central european cohort. European Urology Focus 2023; **7:** 07

222. Chevli KK, Duff M, Walter P et al: Urinary pca3 as a predictor of prostate cancer in a cohort of 3,073 men undergoing initial prostate biopsy. Journal of Urology 2014; **191:** 1743

223. Sanda MG, Feng Z, Howard DH et al: Association between combined tmprss2:Erg and pca3 rna urinary testing and detection of aggressive prostate cancer. JAMA Oncology 2017; **3:** 1085

224. Haese A, de la Taille A, van Poppel H et al: Clinical utility of the pca3 urine assay in european men scheduled for repeat biopsy. European Urology 2008; **54:** 1081

225. Gittelman MC, Hertzman B, Bailen J et al: Pca3 molecular urine test as a predictor of repeat prostate biopsy outcome in men with previous negative biopsies: A prospective multicenter clinical study. Journal of Urology 2013; **190:** 64

226. Haese A, Trooskens G, Steyaert S et al: Multicenter optimization and validation of a 2-gene mrna urine test for detection of clinically significant prostate cancer before initial prostate biopsy. Journal of Urology 2019; **202:** 256

227. Wagaskar VG, Levy M, Ratnani P et al: A selectmdx/magnetic resonance imaging-based nomogram to diagnose prostate cancer. Cancer Reports 2023; **6:** e1668

228. McKiernan J, Noerholm M, Tadigotla V et al: A urine-based exosomal gene expression test stratifies risk of high-grade prostate cancer in men with prior negative prostate biopsy undergoing repeat biopsy. BMC Urology 2020; **20:** 138

229. Tutrone R, Lowentritt B, Neuman B et al: Exodx prostate test as a predictor of outcomes of high-grade prostate cancer - an interim analysis. Prostate Cancer & Prostatic Diseases 2023; **26:** 596

230. Sultan MI, Huynh LM, Kamil S et al: Utility of noninvasive biomarker testing and mri to predict a prostate cancer diagnosis. International Urology & Nephrology 2024; **56:** 539

231. Wang WW, Sorokin I, Aleksic I et al: Expression of small noncoding rnas in urinary exosomes classifies prostate cancer into indolent and aggressive disease. J Urol 2020; **204:** 466

232. Van Neste L, Partin AW, Stewart GD et al: Risk score predicts high-grade prostate cancer in DNA-methylation positive, histopathologically negative biopsies. Prostate 2016; **76:** 1078

233. Waterhouse RL, Jr., Van Neste L, Moses KA et al: Evaluation of an epigenetic assay for predicting repeat prostate biopsy outcome in african american men. Urology 2019; **128:** 62

234. Huynh-Le MP, Fan CC, Karunamuni R et al: Polygenic hazard score is associated with prostate cancer in multi-ethnic populations. Nat Commun 2021; **12:** 1236

235. Pashayan N, Pharoah PD, Schleutker J et al: Reducing overdiagnosis by polygenic risk-stratified screening: Findings from the finnish section of the erspc. British Journal of Cancer 2015; **113:** 1086

236. Benafif S, Ni Raghallaigh H, McGrowder E et al: The barcode1 pilot: A feasibility study of using germline single nucleotide polymorphisms to target prostate cancer screening. BJU Int 2021; **129:** 325

237. Gerstenbluth RE, Seftel AD, Hampel N et al: The accuracy of the increased prostate specific antigen level (greater than or equal to 20 ng./ml.) in predicting prostate cancer: Is biopsy always required? Journal of Urology 2002; **168:** 1990

238. Abraham NE, Mendhiratta N and Taneja SS: Patterns of repeat prostate biopsy in contemporary clinical practice. Journal of Urology 2015; **193:** 1178

239. Bittner N, Merrick GS, Butler WM et al: Incidence and pathological features of prostate cancer detected on transperineal template guided mapping biopsy after negative transrectal ultrasound guided biopsy. Journal of Urology 2013; **190:** 509

240. Giulianelli R, Brunori S, Gentile BC et al: Saturation biopsy technique increase the capacity to diagnose adenocarcinoma of prostate in patients with psa < 10 ng/ml, after a first negative biopsy. Archivio Italiano di Urologia, Andrologia 2011;

83: 154

241. Nam RK, Toi A, Trachtenberg J et al: Variation in patterns of practice in diagnosing screen-detected prostate cancer. BJU International 2004; **94:** 1239

242. Ploussard G, Nicolaiew N, Marchand C et al: Risk of repeat biopsy and prostate cancer detection after an initial extended negative biopsy: Longitudinal follow-up from a prospective trial. BJU International 2013; **111:** 988

243. Thompson IM, Tangen CM, Ankerst DP et al: The performance of prostate specific antigen for predicting prostate cancer is maintained after a prior negative prostate biopsy. J Urol 2008; **180:** 544

244. Cucchiara V, Cooperberg MR, Dall'Era M et al: Genomic markers in prostate cancer decision making. Eur Urol 2018; **73:** 572

245. Foley RW, Maweni RM, Gorman L et al: European randomised study of screening for prostate cancer (erspc) risk calculators significantly outperform the prostate cancer prevention trial (pcpt) 2.0 in the prediction of prostate cancer: A multi-institutional study. BJU International 2016; **1:** 706

246. Kim TS, Ko KJ, Shin SJ et al: Multiple cores of high grade prostatic intraepithelial neoplasia and any core of atypia on first biopsy are significant predictor for cancer detection at a repeat biopsy. Korean Journal of Urology 2015; **56:** 796

247. Palmstedt E, Mansson M, Franlund M et al: Long-term outcomes for men in a prostate screening trial with an initial benign prostate biopsy: A population-based cohort. Eur Urol Oncol 2019; **2:** 716

248. Klemann N, Roder MA, Helgstrand JT et al: Risk of prostate cancer diagnosis and mortality in men with a benign initial transrectal ultrasound-guided biopsy set: A population-based study. Lancet Oncol 2017; **18:** 221

249. Roobol MJ, Steyerberg EW, Kranse R et al: A risk-based strategy improves prostate-specific antigen-driven detection of prostate cancer. Eur Urol 2010; **57:** 79

250. Nam RK, Satkunavisam R, Chin JL et al: Next-generation prostate cancer risk calculator for primary care physicians. Can Urol Assoc J 2018; **12:** E64

251. Nordstrom T, Akre O, Aly M et al: Prostate-specific antigen (psa) density in the diagnostic algorithm of prostate cancer. Prostate Cancer & Prostatic Diseases 2018; **21:** 57

252. Hansen NL, Barrett T, Koo B et al: The influence of prostate-specific antigen density on positive and negative predictive values of multiparametric magnetic resonance imaging to detect gleason score 7-10 prostate cancer in a repeat biopsy setting. BJU International 2017; **119:** 724

253. Tosoian JJ, Sessine MS, Trock BJ et al: Myprostatescore in men considering repeat biopsy: Validation of a simple testing approach. Prostate Cancer & Prostatic Diseases 2023; **26:** 563

254. Stewart GD, Van Neste L, Delvenne P et al: Clinical utility of an epigenetic assay to detect occult prostate cancer in histopathologically negative biopsies: Results of the matloc study. Journal of Urology 2013; **189:** 1110

255. Partin AW, Van Neste L, Klein EA et al: Clinical validation of an epigenetic assay to predict negative histopathological results in repeat prostate biopsies. J Urol 2014; **192:** 1081

256. Tosoian JJ, Singhal U, Davenport MS et al: Urinary myprostatescore (mps) to rule out clinically-significant cancer in men with equivocal (pi-rads 3) multiparametric mri: Addressing an unmet clinical need. Urology 2022; **164:** 184

257. Herawi M, Kahane H, Cavallo C and Epstein JI: Risk of prostate cancer on first re-biopsy within 1 year following a diagnosis of high grade prostatic intraepithelial neoplasia is related to the number of cores sampled. J Urol 2006; **175:** 121

258. Tosoian JJ, Alam R, Ball MW et al: Managing high-grade prostatic intraepithelial neoplasia (hgpin) and atypical glands on prostate biopsy. Nat Rev Urol 2018; **15:** 55

259. Bishara T, Ramnani DM and Epstein JI: High-grade prostatic intraepithelial neoplasia on needle biopsy: Risk of cancer on repeat biopsy related to number of involved cores and morphologic pattern. American Journal of Surgical Pathology 2004; **28:** 629

260. Godoy G, Huang GJ, Patel T and Taneja SS: Long-term follow-up of men with isolated high-grade prostatic intra-epithelial neoplasia followed by serial delayed interval biopsy. Urology 2011; **77:** 669

261. Patel P, Nayak JG, Biljetina Z et al: Prostate cancer after initial high-grade prostatic intraepithelial neoplasia and benign prostate biopsy. Canadian Journal of Urology 2015; **22:** 8056

262. Wiener S, Haddock P, Cusano J et al: Incidence of clinically significant prostate cancer after a diagnosis of atypical small acinar proliferation, high-grade prostatic intraepithelial neoplasia, or benign tissue. Urology 2017; **110:** 161

263. Girasole CR, Cookson MS, Putzi MJ et al: Significance of atypical and suspicious small acinar proliferations, and high grade prostatic intraepithelial neoplasia on prostate biopsy: Implications for cancer detection and biopsy strategy. Journal of Urology 2006; **175:** 929

264. Tan PH, Tan HW, Tan Y et al: Is high-grade prostatic intraepithelial neoplasia on needle biopsy different in an asian population: A clinicopathologic study performed in singapore. Urology 2006; **68:** 800

265. Akhavan A, Keith JD, Bastacky SI et al: The proportion of cores with high-grade prostatic intraepithelial neoplasia on extended-pattern needle biopsy is significantly associated with prostate cancer on site-directed repeat biopsy. BJU Int 2007; **99:** 765

266. Oderda M, Rosazza M, Agnello M et al: Natural history of widespread high grade prostatic intraepithelial neoplasia and atypical small acinar proliferation: Should we rebiopsy them all? Scandinavian Journal of Urology 2021; **55:** 129

267. Netto GJ and Epstein JI: Widespread high-grade prostatic intraepithelial neoplasia on prostatic needle biopsy: A significant likelihood of subsequently diagnosed adenocarcinoma. Am J Surg Pathol 2006; **30:** 1184

268. Morote J, Schwartzmann I, Celma A et al: The current recommendation for the management of isolated high-grade prostatic intraepithelial neoplasia. BJU International 2021; **10:** 10

269. Iczkowski KA, Bassler TJ, Schwob VS et al: Diagnosis of "suspicious for malignancy" in prostate biopsies: Predictive value for cancer. Urology 1998; **51:** 749

270. Iczkowski KA, Chen HM, Yang XJ and Beach RA: Prostate cancer diagnosed after initial biopsy with atypical small acinar proliferation suspicious for malignancy is similar to cancer found on initial biopsy. Urology 2002; **60:** 851

271. Schlesinger C, Bostwick DG and Iczkowski KA: High-grade prostatic intraepithelial neoplasia and atypical small acinar proliferation: Predictive value for cancer in current practice. Am J Surg Pathol 2005; **29:** 1201

272. Epstein JI and Herawi M: Prostate needle biopsies containing prostatic intraepithelial neoplasia or atypical foci suspicious for carcinoma: Implications for patient care. J Urol 2006; **175:** 820

273. Koca O, Caliskan S, Ozturk MI et al: Significance of atypical small acinar proliferation and high-grade prostatic intraepithelial neoplasia in prostate biopsy. Korean Journal of Urology 2011; **52:** 736

274. Leone A, Gershman B, Rotker K et al: Atypical small acinar proliferation (asap): Is a repeat biopsy necessary asap? A multi-institutional review. Prostate Cancer & Prostatic Diseases 2016; **19:** 68

275. Warlick C, Feia K, Tomasini J et al: Rate of gleason 7 or higher prostate cancer on repeat biopsy after a diagnosis of atypical small acinar proliferation. Prostate Cancer & Prostatic Diseases 2015; **18:** 255

276. Dorin RP, Wiener S, Harris CD and Wagner JR: Prostate atypia: Does repeat biopsy detect clinically significant prostate cancer? Prostate 2015; **75:** 673

277. Ji WT, Jin XF and Wang Y: The rate of clinically significant prostate cancer on repeat biopsy after a diagnosis of atypical small acinar proliferation: A systematic review and meta-analysis. Oncology 2024; **102:** 631

278. Shah RB, Magi-Galluzzi C, Han B and Zhou M: Atypical cribriform lesions of the prostate: Relationship to prostatic carcinoma and implication for diagnosis in prostate biopsies. Am J Surg Pathol 2010; **34:** 470

279. Shah RB and Zhou M: Atypical cribriform lesions of the prostate: Clinical significance, differential diagnosis and current concept of intraductal carcinoma of the prostate. Adv Anat Pathol 2012; **19:** 270

280. Shah RB, Yoon J, Liu G and Tian W: Atypical intraductal proliferation and intraductal carcinoma of the prostate on core needle biopsy: A comparative clinicopathological and molecular study with a proposal to expand the morphological spectrum of intraductal carcinoma. Histopathology 2017; **71:** 693

281. Hickman RA, Yu H, Li J et al: Atypical intraductal cribriform proliferations of the prostate exhibit similar molecular clinicopathologic characteristics as intraductal carcinoma of the prostate. Am J Surg Pathol 2017; **41:** 550

282. Shah RB, Nguyen JK, Przybycin CG et al: Atypical intraductal proliferation detected in prostate needle biopsy is a marker of unsampled intraductal carcinoma and other adverse pathological features: A prospective clinicopathological study of 62 cases with emphasis on pathological outcomes. Histopathology 2019; **75(3):** 346

283. Hoeks CM, Schouten MG, Bomers JG et al: Three-tesla magnetic resonance-guided prostate biopsy in men with increased prostate-specific antigen and repeated, negative, random, systematic, transrectal ultrasound biopsies: Detection of clinically significant prostate cancers. European Urology 2012; **62:** 902

284. Arsov C, Rabenalt R, Blondin D et al: Prospective randomized trial comparing magnetic resonance imaging (mri)-guided in-bore biopsy to mri-ultrasound fusion and transrectal ultrasound-guided prostate biopsy in patients with prior negative biopsies. European Urology 2015; **68:** 713

285. Hambrock T, Somford DM, Hoeks C et al: Magnetic resonance imaging guided prostate biopsy in men with repeat negative biopsies and increased prostate specific antigen. Journal of Urology 2010; **183:** 520

286. Sonn GA, Chang E, Natarajan S et al: Value of targeted prostate biopsy using magnetic resonance-ultrasound fusion in men with prior negative biopsy and elevated prostate-specific antigen. European Urology 2014; **65:** 809

287. Meermeier NP, Foster BR, Liu JJ et al: Impact of direct mri-guided biopsy of the prostate on clinical management. AJR American Journal of Roentgenology 2019; **213:** 371

288. Da Rosa MR, Milot L, Sugar L et al: A prospective comparison of mri-us fused targeted biopsy versus systematic ultrasound-guided biopsy for detecting clinically significant prostate cancer in patients on active surveillance. J Magn Reson Imaging 2015; **41:** 220

289. Elkjær MC, Andersen MH, Høyer S et al: Prostate cancer: In-bore magnetic resonance guided biopsies at active surveillance inclusion improve selection of patients for active treatment. Acta Radiol 2018; **59:** 619

290. Welch HG, Fisher ES, Gottlieb DJ and Barry MJ: Detection of prostate cancer via biopsy in the medicare-seer population during the psa era. Journal of the National Cancer Institute 2007; **99:** 1395

291. Distler FA, Radtke JP, Bonekamp D et al: The value of psa density in combination with pi-rads for the accuracy of prostate cancer prediction. J Urol 2017; **198:** 575

292. Druskin SC, Tosoian JJ, Young A et al: Combining prostate health index density, magnetic resonance imaging and prior negative biopsy status to improve the detection of clinically significant prostate cancer. BJU International 2018; **121:** 619

293. Kamal O, Comerford J, Foster BR et al: Intermediate-term oncological outcomes after a negative endorectal coil multiparametric mri of the prostate in patients without biopsy proven prostate cancer. Clinical Imaging 2022; **92:** 112

294. Zhen L, Liu X, Yegang C et al: Accuracy of multiparametric magnetic resonance imaging for diagnosing prostate cancer: A systematic review and meta-analysis. BMC Cancer 2019; **19:** 1244

295. Patel N, Cricco-Lizza E, Kasabwala K et al: The role of systematic and targeted biopsies in light of overlap on magnetic resonance imaging ultrasound fusion biopsy. European Urology Oncology 2018; **1:** 263

296. Salami SS, Ben-Levi E, Yaskiv O et al: In patients with a previous negative prostate biopsy and a suspicious lesion on magnetic resonance imaging, is a 12-core biopsy still necessary in addition to a targeted biopsy? BJU International 2015; **115:** 562

297. Loeb S, Carter HB, Berndt SI et al: Is repeat prostate biopsy associated with a greater risk of hospitalization? Data from seer-medicare. Journal of Urology 2013; **189:** 867

298. Fujita K, Landis P, McNeil BK and Pavlovich CP: Serial prostate biopsies are associated with an increased risk of erectile dysfunction in men with prostate cancer on active surveillance. J Urol 2009; **182:** 2664

299. Hale GR, Czarniecki M, Cheng A et al: Comparison of elastic and rigid registration during magnetic resonance imaging/ultrasound fusion-guided prostate biopsy: A multi-operator phantom study. J Urol 2018; **200:** 1114

300. Izadpanahi MH, Elahian A, Gholipour F et al: Diagnostic yield of fusion magnetic resonance-guided prostate biopsy versus cognitive-guided biopsy in biopsy-naive patients: A head-to-head randomized controlled trial. Prostate Ca Prostatic Diseases 2021; **27:** 27

301. Watts KL, Frechette L, Muller B et al: Systematic review and meta-analysis comparing cognitive vs. Image-guided fusion prostate biopsy for the detection of prostate cancer. Urologic Oncology 2020; **38:** 734.e19

302. Hamid S, Donaldson IA, Hu Y et al: The smarttarget biopsy trial: A prospective, within-person randomised, blinded trial comparing the accuracy of visual-registration and magnetic resonance imaging/ultrasound image-fusion targeted biopsies for prostate cancer risk stratification. European Urology 2019; **75:** 733

303. Liang L, Cheng Y, Qi F et al: A comparative study of prostate cancer detection rate between transperineal cog-tb and transperineal fus-tb in patients with psa <=20 ng/ml. Journal of Endourology 2020; **34:** 1008

304. Dai Z, Liu Y, Huangfu Z et al: Magnetic resonance imaging (mri)-targeted biopsy in patients with prostate-specific antigen (psa) levels <20 ng/ml: A single-center study in northeastern china. Medical Science Monitor 2021; **27:** e930234

305. Wegelin O, Exterkate L, van der Leest M et al: The future trial: A multicenter randomised controlled trial on target biopsy techniques based on magnetic resonance imaging in the diagnosis of prostate cancer in patients with prior negative biopsies. European Urology 2019; **75:** 582

306. Sonmez G, Demirtas T, Tombul ST et al: What is the ideal number of biopsy cores per lesion in targeted prostate biopsy? Prostate International 2020; **8:** 112

307. Subramanian N, Recchimuzzi DZ, Xi Y et al: Impact of the number of cores on the prostate cancer detection rate in men undergoing in-bore magnetic resonance imaging-guided targeted biopsies. J Comput Assist Tomogr 2021; **45:** 203

308. Tu X, Lin T, Cai D et al: The optimal core number and site for mri-targeted biopsy of prostate? A systematic review and pooled analysis. Minerva Urologica e Nefrologica 2020; **72:** 144

309. Eastham JA, Auffenberg GB, Barocas DA et al: Clinically localized prostate cancer: Aua/astro guideline, part i: Introduction, risk assessment, staging, and risk-based management. J Urol 2022; **208:** 10

310. Scott S, Samaratunga H, Chabert C et al: Is transperineal prostate biopsy more accurate than transrectal biopsy in determining final gleason score and clinical risk category? A comparative analysis. BJU Int 2015; **116 Suppl 3:** 26

311. Uleri A, Baboudjian M, Tedde A et al: Is there an impact of transperineal versus transrectal magnetic resonance imaging-targeted biopsy in clinically significant prostate cancer detection rate? A systematic review and meta-analysis. Eur Urol Oncol 2023; **6:** 621

312. Diamand R, Guenzel K, Mjaess G et al: Transperineal or transrectal magnetic resonance imaging-targeted biopsy for prostate cancer detection. Eur Urol Focus 2024; **10:** 805

313. Kaneko M, Medina LG, Lenon MSL et al: Transperineal vs transrectal magnetic resonance and ultrasound image fusion prostate biopsy: A pair-matched comparison. Sci Rep 2023; **13:** 13457

314. Zattoni F, Rajwa P, Miszczyk M et al: Transperineal versus transrectal magnetic resonance imaging-targeted prostate biopsy: A systematic review and meta-analysis of prospective studies. Eur Urol Oncol 2024; **7:** 1303

315. Hara R, Jo Y, Fujii T et al: Optimal approach for prostate cancer detection as initial biopsy: Prospective randomized study comparing transperineal versus transrectal systematic 12-core biopsy. Urology 2008; **71:** 191

316. Ploussard G, Barret E, Fiard G et al: Transperineal versus transrectal magnetic resonance imaging-targeted biopsies for prostate cancer diagnosis: Final results of the randomized perfect trial (ccafu-pr1). Eur Urol Oncol 2024; **7:** 1080

317. Bryant RJ, Marian IR, Williams R et al: Local anaesthetic transperineal biopsy versus transrectal prostate biopsy in prostate cancer detection (translate): A multicentre, randomised, controlled trial. Lancet Oncol 2025; **26:** 583

318. Mian BM, Feustel PJ, Aziz A et al: Complications following transrectal and transperineal prostate biopsy: Results of the probe-pc randomized clinical trial. J Urol 2024; **211:** 205

319. Mian BM, Feustel PJ, Aziz A et al: Clinically significant prostate cancer detection following transrectal and transperineal biopsy: Results of the prostate biopsy efficacy and complications randomized clinical trial. J Urol 2024; **212:** 21

320. Hu JC, Assel M, Allaf ME et al: Transperineal versus transrectal magnetic resonance imaging-targeted and systematic prostate biopsy to prevent infectious complications: The prevent randomized trial. Eur Urol 2024; **86:** 61

321. Berg S, Tully KH, Hoffmann V et al: Assessment of complications after transperineal and transrectal prostate biopsy using a risk-stratified pathway identifying patients at risk for post-biopsy infections. Scand J Urol 2023; **57:** 41

322. Berquin C, Perletti G, Develtere D et al: Transperineal vs. Transrectal prostate biopsies under local anesthesia: A prospective cohort study on patient tolerability and complication rates. Urol Oncol 2023; **41:** 388.e17

323. Hogenhout R, Remmers S, van Leenders G and Roobol MJ: The transition from transrectal to transperineal prostate biopsy without antibiotic prophylaxis: Cancer detection rates and complication rates. Prostate Cancer Prostatic Dis 2023; **26:** 581

324. Meyer AR, Mamawala M, Winoker JS et al: Transperineal prostate biopsy improves the detection of clinically significant prostate cancer among men on active surveillance. J Urol 2021; **205:** 1069

325. Roberts MJ, Bennett HY, Harris PN et al: Prostate biopsy-related infection: A systematic review of risk factors, prevention strategies, and management approaches. Urology 2017; **104:** 11

326. Womble PR, Linsell SM, Gao Y et al: A statewide intervention to reduce hospitalizations after prostate biopsy. J Urol 2015; **194:** 403

327. Drazer MW, Prasad SM, Huo D et al: National trends in prostate cancer screening among older american men with limited 9-year life expectancies: Evidence of an increased need for shared decision making. Cancer 2014; **120:** 1491

328. Nik-Ahd F, Jarjour A, Figueiredo J et al: Prostate-specific antigen screening in transgender patients. Eur Urol 2023; **83:** 48

329. Kinnaird A, Luger F, Cash H et al: Microultrasonography-guided vs mri-guided biopsy for prostate cancer diagnosis: The optimum randomized clinical trial. Jama 2025; **333:** 1679

330. Ng A, Asif A, Agarwal R et al: Biparametric vs multiparametric mri for prostate cancer diagnosis: The prime diagnostic clinical trial. Jama 2025; **334:** 1170

American Urological Association

1000 Corporate Boulevard
Linthicum, MD 21090
Phone: 410-689-3700
Toll-Free: 1-800-828-7866
Fax: 410-689-3912
Email: aua@AUAnet.org